

# SOLIDWORKS API (.NET VBA C#) — Reference (13/337): IAssemblyDoc (part 1 of 5)

SOLIDWORKS API Documentation (converted from CHM)

# IAssemblyDoc (part 1 of 5) (Type)

## IAssemblyDoc Interface

[See Also](#) [Members](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) : IAssemblyDoc Interface

Allows access to functions that perform assembly operations; for example, adding new components, adding mate conditions, hiding, and exploding components.

**NOTE:** Click the **Members** link, located near the top of the topic, to see this interface's methods and properties.

### .NET Syntax

Visual Basic (Declaration)

```
Public Interface IAssemblyDoc
```

Visual Basic (Usage)

```
Dim instance As IAssemblyDoc
```

C#

```
public interface IAssemblyDoc
```

C++/CLI

```
public interface class IAssemblyDoc
```

### Visual Basic for Applications (VBA) Syntax

See [AssemblyDoc](#).

### Example

[Run Interference Detection \(C#\)](#)  
[Run Interference Detection \(VB.NET\)](#)  
[Run Interference Detection \(VBA\)](#)  
[Insert and Save Virtual Assembly \(C#\)](#)  
[Insert and Save Virtual Assembly \(VB.NET\)](#)  
[Insert and Save Virtual Assembly \(VBA\)](#)  
[Add Components \(C#\)](#)  
[Add Components \(VB.NET\)](#)  
[Add Components \(VBA\)](#)  
[Add Component and Mate \(C++\)](#)  
[Add Component and Mate \(VBA\)](#)  
[Insert MidSurface in Assembly \(C#\)](#)  
[Insert MidSurface in Component \(VB.NET\)](#)  
[Insert MidSurface in Component \(VBA\)](#)  
[Create Auto Route \(VBA\)](#)  
[Create Auto Route \(VB.NET\)](#)  
[Create Auto Route \(C#\)](#)  
[Add Component and Mate \(VB.NET\)](#)  
[Add Component and Mate Example \(C#\)](#)

### Remarks

The SOLIDWORKS API includes functions that are common to all document types; for example, determining the file name associated with a document is a common operation. To expose common document-level functions, the SOLIDWORKS API uses the [IModelDoc2](#) object.

Events are implemented with delegates in the Microsoft .NET Framework. See the [Overview](#) topic for a list of delegates for this interface.

### Accessors

[ICollisionDetectionManager::GetAssembly](#)

### Access Diagram

[AssemblyDoc](#)

### See Also

## IAssemblyDoc Interface Methods

[See Also](#)

[SolidWorks.Interop.sldworks Namespace](#) : IAssemblyDoc Interface

For a list of all members of this type, see [IAssemblyDoc members](#).

### Public Methods

| Name                                                                                                                             | Description                                                                                                                                                                                   |
|----------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  <a href="#">ActivateGroundPlane</a>            | Activates the ground plane for the specified configurations.                                                                                                                                  |
|  <a href="#">AddComponent</a>                   | Obsolete. Superseded by <a href="#">IAssemblyDoc::AddComponent4</a> .                                                                                                                         |
|  <a href="#">AddComponent2</a>                  | Obsolete. Superseded by <a href="#">IAssemblyDoc::AddComponent4</a> .                                                                                                                         |
|  <a href="#">AddComponent4</a>                  | Obsolete. Superseded by <a href="#">IAssemblyDoc::AddComponent5</a> .                                                                                                                         |
|  <a href="#">AddComponent5</a>                  | Adds the specified component for the specified configuration options to this assembly.                                                                                                        |
|  <a href="#">AddComponentConfiguration</a>      | Adds a new configuration for the last selected assembly component.                                                                                                                            |
|  <a href="#">AddComponents</a>                  | Obsolete. Superseded by <a href="#">IAssemblyDoc::AddComponents3</a> .                                                                                                                        |
|  <a href="#">AddComponents3</a>                 | Adds multiple components to the assembly.                                                                                                                                                     |
|  <a href="#">AddConcentricMateWithTolerance</a> | Adds a misaligned concentric mate to this assembly.                                                                                                                                           |
|  <a href="#">AddDistanceMate</a>                | Adds a distance mate to this assembly.                                                                                                                                                        |
|  <a href="#">AddMate</a>                        | Obsolete. Superseded by <a href="#">IAssemblyDoc::AddMate3</a> .                                                                                                                              |
|  <a href="#">AddMate2</a>                       | Obsolete. Superseded by <a href="#">IAssemblyDoc::AddMate3</a> .                                                                                                                              |
|  <a href="#">AddMate3</a>                       | Obsolete. Superseded by <a href="#">IAssemblyDoc::AddMate4</a> .                                                                                                                              |
|  <a href="#">AddMate4</a>                     | Obsolete. Superseded by <a href="#">IAssemblyDoc::AddMate5</a> .                                                                                                                              |
|  <a href="#">AddMate5</a>                     | Obsolete. Superseded by <a href="#">IAssemblyDoc::CreateMate</a> .                                                                                                                            |
|  <a href="#">AddPipePenetration</a>           | Penetrates the adjacent fitting or pipe with the pipe that ends at the selected sketch point.                                                                                                 |
|  <a href="#">AddPipingFitting</a>             | Adds a pipe fitting to the current piping assembly.                                                                                                                                           |
|  <a href="#">AddPLMComponent</a>              | Adds the specified component from a 3DEXPERIENCE collaborative space to the specified location in this assembly in <a href="#">SOLIDWORKS Design connected to the 3DEXPERIENCE platform</a> . |
|  <a href="#">AddSmartComponent</a>            | Adds the specified component at the specified coordinates as a Smart Component to this assembly.                                                                                              |
|  <a href="#">AddToFeatureScope</a>            | Adds a component to the scope of the currently selected assembly feature.                                                                                                                     |
|  <a href="#">AssemblyPartToggle</a>           | Obsolete. Superseded by <a href="#">IAssemblyDoc::EditAssembly</a> and <a href="#">IAssemblyDoc::EditPart2</a> .                                                                              |
|  <a href="#">AutoAngleAxis</a>                | Automatically detect the axis for an angle mate.                                                                                                                                              |
|  <a href="#">AutoExplode</a>                  | Automatically generates an exploded view of the current assembly configuration.                                                                                                               |
|  <a href="#">AutoMateRepair</a>               | Automatically repairs broken mates in this assembly.                                                                                                                                          |
|  <a href="#">CollectAllSupportiveMates</a>    | Gets all mates supportive of a mate controller in this assembly.                                                                                                                              |
|  <a href="#">CompConfigProperties</a>         | Obsolete. Superseded by <a href="#">IAssemblyDoc::CompConfigProperties4</a> .                                                                                                                 |
|  <a href="#">CompConfigProperties2</a>        | Obsolete. Superseded by <a href="#">IAssemblyDoc::CompConfigProperties4</a> .                                                                                                                 |
|  <a href="#">CompConfigProperties3</a>        | Obsolete. Superseded by <a href="#">IAssemblyDoc::CompConfigProperties4</a> .                                                                                                                 |
|  <a href="#">CompConfigProperties4</a>        | Obsolete. Superseded by <a href="#">IAssemblyDoc::CompConfigProperties5</a> .                                                                                                                 |
|  <a href="#">CompConfigProperties5</a>        | Obsolete. Superseded by <a href="#">IAssemblyDoc::CompConfigProperties6</a> .                                                                                                                 |
|  <a href="#">CompConfigProperties6</a>        | Sets the properties for the selected <a href="#">component</a> in the specified <a href="#">configuration</a> .                                                                               |
|  <a href="#">ComponentReload</a>              | Obsolete. Superseded by <a href="#">IAssemblyDoc::ComponentReload2</a> .                                                                                                                      |
|  <a href="#">ComponentReload2</a>             | Reloads and/or sets the read-only state of the specified component.                                                                                                                           |
|  <a href="#">CopyWithMates</a>                | Obsolete. Superseded by <a href="#">IAssemblyDoc::CopyWithMates2</a> .                                                                                                                        |
|  <a href="#">CopyWithMates2</a>               | Copies one or more components with mates in this assembly.                                                                                                                                    |

|                                                                                     |                                                  |                                                                                                                      |
|-------------------------------------------------------------------------------------|--------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
|    |                                                  |                                                                                                                      |
|    | <a href="#">CreateExplodedView</a>               | Creates an explode view of the active assembly configuration.                                                        |
|    | <a href="#">CreateMate</a>                       | Creates a mate with the specified feature data object.                                                               |
|    | <a href="#">CreateMateData</a>                   | Creates a mate feature data object for the specified mate type.                                                      |
|    | <a href="#">CreateSmartComponent</a>             | Creates a Smart Component.                                                                                           |
|    | <a href="#">CreateSpeedPak</a>                   | Creates the specified type of SpeedPak for the active configurations of the selected subassemblies in this assembly. |
|    | <a href="#">DeleteSelections</a>                 | Delete either the selected components of a subassembly or the subassembly of the selected component.                 |
|    | <a href="#">DissolveComponentPattern</a>         | Dissolves the selected component patterns.                                                                           |
|    | <a href="#">DissolveSubAssembly</a>              | Dissolves the selected subassembly in this assembly.                                                                 |
|    | <a href="#">EditAssembly</a>                     | Switches back to the assembly document for editing.                                                                  |
|    | <a href="#">EditConcentricMate</a>               | Edits a misaligned concentric mate.                                                                                  |
|    | <a href="#">EditDistanceMate</a>                 | Edits a distance mate.                                                                                               |
|    | <a href="#">EditMate</a>                         | Obsolete. Superseded by <a href="#">IAssemblyDoc.EditMate2</a> .                                                     |
|    | <a href="#">EditMate2</a>                        | Obsolete. Superseded by <a href="#">IAssemblyDoc.EditMate3</a> .                                                     |
|    | <a href="#">EditMate3</a>                        | Obsolete. Superseded by <a href="#">IAssemblyDoc.EditMate4</a> .                                                     |
|    | <a href="#">EditMate4</a>                        | Obsolete. See the Remarks in each standard, advanced, and mechanical mate's feature data interface.                  |
|    | <a href="#">EditPart</a>                         | Obsolete. Superseded by <a href="#">IAssemblyDoc.EditPart2</a> .                                                     |
|    | <a href="#">EditPart2</a>                        | Edits the selected part within the context of an assembly.                                                           |
|    | <a href="#">EditRebuild</a>                      | Obsolete. Superseded by <a href="#">IModelDoc2.EditRebuild3</a> .                                                    |
|    | <a href="#">ExitIsolate</a>                      | Exits <a href="#">isolating the selected components</a> and returns the assembly to its original display state.      |
|    | <a href="#">FeatureByName</a>                    | Returns the <a href="#">IFeature</a> object for the named feature in the assembly.                                   |
|   | <a href="#">FeatureExtrusion</a>                 | Obsolete. Superseded by <a href="#">IFeatureManager.FeatureExtrusion</a> .                                           |
|  | <a href="#">FileDeriveComponentPart</a>          | Creates a new part document from the currently selected assembly component.                                          |
|  | <a href="#">FixComponent</a>                     | Fixes the selected component; i.e., makes it immovable.                                                              |
|  | <a href="#">ForceRebuild</a>                     | Obsolete. Superseded by <a href="#">IModelDoc2.ForceRebuild3</a> .                                                   |
|  | <a href="#">ForceRebuild2</a>                    | Obsolete. Superseded by <a href="#">IModelDoc2.ForceRebuild3</a> .                                                   |
|  | <a href="#">ForceUpdateElectricalData</a>        | Obsolete. Superseded by <a href="#">IAssemblyDoc.ForceUpdateElectricalData2</a> .                                    |
|  | <a href="#">ForceUpdateElectricalData2</a>       | Forces an update of electrical data.                                                                                 |
|  | <a href="#">GetActiveGroundPlane</a>             | Gets the active ground plane for the specified configurations.                                                       |
|  | <a href="#">GetAdvancedSelection</a>             | Gets the advanced component selection criteria object for this assembly.                                             |
|  | <a href="#">GetBox</a>                           | Gets the bounding box.                                                                                               |
|  | <a href="#">GetComponentByID</a>                 | Gets a top-level assembly component using its component ID.                                                          |
|  | <a href="#">GetComponentByName</a>               | Gets the specified top-level assembly component.                                                                     |
|  | <a href="#">GetComponentCount</a>                | Gets the number of components in the active configuration of this assembly.                                          |
|  | <a href="#">GetComponents</a>                    | Gets all of the components in the active configuration of this assembly.                                             |
|  | <a href="#">GetDragOperator</a>                  | Gets the drag operator for dynamic drag operations in this assembly.                                                 |
|  | <a href="#">GetDroppedAIEntity</a>               | Gets a pointer to the entity where a file is dropped into this assembly.                                             |
|  | <a href="#">GetEditTarget</a>                    | Gets the model document that is currently being edited.                                                              |
|  | <a href="#">GetEditTargetComponent</a>           | Gets the component that is currently being edited.                                                                   |
|  | <a href="#">GetExplodedViewConfigurationName</a> | Gets the name of the configuration for the specified exploded view.                                                  |
|  | <a href="#">GetExplodedViewCount</a>             | Obsolete. Superseded by <a href="#">IAssemblyDoc.GetExplodedViewCount2</a> .                                         |
|  | <a href="#">GetExplodedViewCount2</a>            | Gets the number of exploded views in the specified configuration.                                                    |
|  | <a href="#">GetExplodedViewNames</a>             | Obsolete. Superseded by <a href="#">IAssemblyDoc.GetExplodedViewNames2</a> .                                         |

|                                                                                     |                                                 |                                                                                                                                |
|-------------------------------------------------------------------------------------|-------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
|    |                                                 |                                                                                                                                |
|    | <a href="#">GetExplodedViewNames2</a>           | Gets the names of the exploded views in the specified configuration.                                                           |
|    | <a href="#">GetFeatureScope</a>                 | Gets the components affected by this feature.                                                                                  |
|    | <a href="#">GetFeatureScopeCount</a>            | Gets the number of components affected by this feature.                                                                        |
|    | <a href="#">GetFirstMember</a>                  | Obsolete. Superseded by <a href="#">IComponent2::GetChildren</a> .                                                             |
|    | <a href="#">GetLargeDesignReviewState</a>       | Gets the Large Design Review state of this assembly.                                                                           |
|    | <a href="#">GetLightWeightComponentCount</a>    | Gets the number of lightweight components in the assembly.                                                                     |
|    | <a href="#">GetPhysicalSimulationComponents</a> | Obsolete. Not superseded.                                                                                                      |
|    | <a href="#">GetRouteManager</a>                 | Gets the SOLIDWORKS Routing API.                                                                                               |
|    | <a href="#">GetSelectedMember</a>               | Obsolete. Superseded by <a href="#">ISelectionMgr::GetSelectedObject6</a> .                                                    |
|    | <a href="#">GetSimulation</a>                   | Obsolete. Not superseded.                                                                                                      |
|    | <a href="#">GetUnloadedComponentNames</a>       | Gets the unloaded components' paths, referenced configuration names, reasons why they are unloaded, document types, and names. |
|    | <a href="#">GetVisibleComponentsInView</a>      | Gets a list of visible components in this assembly to save as solid bodies.                                                    |
|    | <a href="#">GetVisibleComponentsInViewCount</a> | Gets the number of visible components in this assembly.                                                                        |
|    | <a href="#">HasUnloadedComponents</a>           | Gets whether this assembly has hidden or suppressed unloaded components.                                                       |
|    | <a href="#">HideComponent</a>                   | Obsolete. Superseded by <a href="#">IModelDoc2::HideComponent2</a> .                                                           |
|    | <a href="#">IAddComponent2</a>                  | Obsolete. Superseded by <a href="#">IAssemblyDoc::AddComponent4</a> .                                                          |
|    | <a href="#">IAddComponent3</a>                  | Obsolete. Superseded by <a href="#">IAssemblyDoc::AddComponent4</a> .                                                          |
|    | <a href="#">IAddComponents</a>                  | Obsolete. Superseded by <a href="#">IAssemblyDoc::IAddComponents2</a> .                                                        |
|    | <a href="#">IAddComponents2</a>                 | Obsolete. Superseded by <a href="#">IAssemblyDoc::IAddComponents3</a> .                                                        |
|    | <a href="#">IAddComponents3</a>                 | Adds multiple components to the assembly.                                                                                      |
|   | <a href="#">IComponentReload2</a>               | Obsolete. Superseded by <a href="#">IAssemblyDoc::ReplaceComponents</a> .                                                      |
|  | <a href="#">IComponentReload3</a>               | Obsolete. Superseded by <a href="#">IAssemblyDoc::ReplaceComponents</a> .                                                      |
|  | <a href="#">IFeatureByName</a>                  | Returns the <a href="#">IFeature</a> object for the named feature in the assembly.                                             |
|  | <a href="#">IGetBox</a>                         | Gets the bounding box.                                                                                                         |
|  | <a href="#">IGetComponents</a>                  | Gets all of the components in the active configuration of this assembly.                                                       |
|  | <a href="#">IGetDragOperator</a>                | Gets the drag operator for dynamic drag operations in this assembly.                                                           |
|  | <a href="#">IGetEditTarget</a>                  | Obsolete. Superseded by <a href="#">IAssemblyDoc::IGetEditTarget2</a> .                                                        |
|  | <a href="#">IGetEditTarget2</a>                 | Gets the model document that is currently being edited.                                                                        |
|  | <a href="#">IGetExplodedViewNames</a>           | Obsolete. Superseded by <a href="#">IAssemblyDoc::GetExplodedViewNames2</a> .                                                  |
|  | <a href="#">IGetFeatureScope</a>                | Gets the components affected by this feature.                                                                                  |
|  | <a href="#">IGetFirstMember</a>                 | Obsolete. Superseded by <a href="#">IComponent2::GetChildren</a> .                                                             |
|  | <a href="#">IGetSelectedMember</a>              | Obsolete. Superseded by <a href="#">ISelectionMgr::GetSelectedObject6</a> .                                                    |
|  | <a href="#">IGetVisibleComponentsInView</a>     | Gets a list of visible components in this assembly to save as solid bodies.                                                    |
|  | <a href="#">IMirrorComponents</a>               | Obsolete. Superseded by <a href="#">IAssemblyDoc::MirrorComponents2</a> .                                                      |
|  | <a href="#">InsertCavity</a>                    | Obsolete. Superseded by <a href="#">IAssemblyDoc::InsertCavity4</a> .                                                          |
|  | <a href="#">InsertCavity2</a>                   | Obsolete. Superseded by <a href="#">IAssemblyDoc::InsertCavity4</a> .                                                          |
|  | <a href="#">InsertCavity3</a>                   | Obsolete. Superseded by <a href="#">IAssemblyDoc::InsertCavity4</a> .                                                          |
|  | <a href="#">InsertCavity4</a>                   | Inserts a cavity to the active part using a selected component.                                                                |
|  | <a href="#">InsertDerivedPattern</a>            | Obsolete. See <a href="#">IFeatureManager::CreateFeature</a> and the Remarks in <a href="#">IDerivedPatternFeatureData</a> .   |
|  | <a href="#">InsertEnvelope</a>                  | Adds an envelope in the specified configuration name in this assembly.                                                         |
|  | <a href="#">InsertImportedComponent</a>         | Inserts a third-party native CAD part or assembly into the current configuration of this assembly.                             |

|                                                                                     |                                                       |                                                                                                                                                                                               |
|-------------------------------------------------------------------------------------|-------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|    | <a href="#">InsertJoin</a>                            | Obsolete. Superseded by <a href="#">IAssemblyDoc::InsertJoin2</a> .                                                                                                                           |
|    | <a href="#">InsertJoin2</a>                           | Constructs a feature from merged selected components.                                                                                                                                         |
|    | <a href="#">InsertLoadReference</a>                   | Creates a mate load reference to the specified or selected mate.                                                                                                                              |
|    | <a href="#">InsertNewAssembly</a>                     | Creates a new virtual sub-assembly and optionally saves it to the specified file.                                                                                                             |
|    | <a href="#">InsertNewPart</a>                         | Obsolete. Superseded by <a href="#">IAssemblyDoc::InsertNewPart2</a> .                                                                                                                        |
|    | <a href="#">InsertNewPart2</a>                        | Inserts a new part on the specified face or plane.                                                                                                                                            |
|    | <a href="#">InsertNewVirtualAssembly</a>              | Creates a new assembly from this assembly and saves it internally as a virtual component.                                                                                                     |
|    | <a href="#">InsertNewVirtualPart</a>                  | Creates a new part in the context of an assembly and saves the part internally in the assembly file as a virtual component.                                                                   |
|    | <a href="#">InsertWeld</a>                            | Obsolete. Do not use. Superseded by <a href="#">IFeatureManager::InsertCosmeticWeldBead</a> .                                                                                                 |
|    | <a href="#">InsertWeld2</a>                           | Obsolete. Do not use. Superseded by <a href="#">IFeatureManager::InsertCosmeticWeldBead</a> .                                                                                                 |
|    | <a href="#">IReorderComponents</a>                    | Moves components to a different location in the FeatureManager tree.                                                                                                                          |
|    | <a href="#">IReorganizeComponents</a>                 | Reorganizes an assembly's structure by moving the selected components to the selected assembly or sub-assembly.                                                                               |
|    | <a href="#">IsComponentTreeValid</a>                  | Checks the validity of the component tree for this assembly.                                                                                                                                  |
|    | <a href="#">ISetComponentState</a>                    | Sets the suppression state for the specified components.                                                                                                                                      |
|    | <a href="#">ISetComponentVisibility</a>               | Hides or shows the selected component in the specified configurations in this assembly document.                                                                                              |
|    | <a href="#">Isolate</a>                               | Isolates the selected components.                                                                                                                                                             |
|    | <a href="#">IsRouteAssembly</a>                       | Gets whether the assembly document is a routing assembly.                                                                                                                                     |
|    | <a href="#">IToolsCheckInterference2</a>              | Obsolete. See <a href="#">IAssemblyDoc::IToolsCheckInterference3</a> .                                                                                                                        |
|    | <a href="#">IToolsCheckInterference3</a>              | Obsolete.                                                                                                                                                                                     |
|    | <a href="#">LightweightAllResolved</a>                | Sets to lightweight all resolved child components of the selected components.                                                                                                                 |
|    | <a href="#">MakeAssemblyFromSelectedComponents</a>    | Creates a new assembly comprised of the selected components of this assembly.                                                                                                                 |
|   | <a href="#">MakeIndependent</a>                       | Makes the selected component independent.                                                                                                                                                     |
|  | <a href="#">MakeLightWeight</a>                       | Sets the selected components to lightweight.                                                                                                                                                  |
|  | <a href="#">MirrorComponents</a>                      | Obsolete. Superseded by <a href="#">IAssemblyDoc::MirrorComponents2</a> .                                                                                                                     |
|  | <a href="#">MirrorComponents2</a>                     | Obsolete. Superseded by <a href="#">IAssemblyDoc::MirrorComponents3</a> .                                                                                                                     |
|  | <a href="#">MirrorComponents3</a>                     | Obsolete. Superseded by <a href="#">IFeatureManager::CreateDefinition</a> , <a href="#">IFeatureManager::CreateFeature</a> , and <a href="#">IMirrorComponentFeatureData</a> .                |
|  | <a href="#">OpenCompFile</a>                          | Obsolete. Superseded by <a href="#">ISldWorks::ActivateDoc2</a> and <a href="#">ISldWorks::OpenDoc6</a> .                                                                                     |
|  | <a href="#">RemoveFromFeatureScope</a>                | Removes a component from the scope of the currently selected assembly feature.                                                                                                                |
|  | <a href="#">ReorderComponents</a>                     | Moves components to a different location in the FeatureManager design tree.                                                                                                                   |
|  | <a href="#">ReorganizeComponents</a>                  | Reorganizes an assembly's structure by moving the selected components to the selected assembly or sub-assembly.                                                                               |
|  | <a href="#">ReplaceComponents</a>                     | Obsolete. Superseded by <a href="#">IAssemblyDoc::ReplaceComponents2</a> .                                                                                                                    |
|  | <a href="#">ReplaceComponents2</a>                    | Replaces one or more selected components with another model.                                                                                                                                  |
|  | <a href="#">ReplacePLMComponents</a>                  | Replaces a selected <a href="#">SOLIDWORKS Design connected to the 3DEXPERIENCE platform</a> component in this assembly with the specified component from a 3DEXPERIENCE collaborative space. |
|  | <a href="#">ResolveAllLightweight</a>                 | Resolves all lightweight child components of the selected components                                                                                                                          |
|  | <a href="#">ResolveAllLightWeightComponents</a>       | Resolves the lightweight components in this assembly.                                                                                                                                         |
|  | <a href="#">ResolveOutOfDateLightWeightComponents</a> | Resolves the selected out-of-date lightweight component, and any out-of-date lightweight sub-components, in the assembly.                                                                     |
|  | <a href="#">RotateComponent</a>                       | Displays the Rotate Component PropertyManager page.                                                                                                                                           |
|  | <a href="#">RotateComponentAxis</a>                   | Rotates the component axis by a fixed amount.                                                                                                                                                 |
|  | <a href="#">SaveIsolate</a>                           | Saves the display characteristics of the <a href="#">isolated components</a> to a new display state.                                                                                          |
|  | <a href="#">SelectComponentsBySize</a>                | Selects assembly components by percent of assembly size.                                                                                                                                      |
|  | <a href="#">SelectiveOpen</a>                         | Selectively opens the components of an assembly that is opened in Large Design Review mode.                                                                                                   |
|  | <a href="#">SetComponentState</a>                     | Sets the suppression state for the specified components.                                                                                                                                      |
|  | <a href="#">SetComponentSuppression</a>               | Suppresses, resolves, or sets to lightweight selected components of this assembly in the active configuration only.                                                                           |

|                                                                                     |                                                  |                                                                                                                                                                     |
|-------------------------------------------------------------------------------------|--------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|     |                                                  |                                                                                                                                                                     |
|    | <a href="#">SetComponentTransparent</a>          | Enables or disables transparency on the selected components.                                                                                                        |
|    | <a href="#">SetComponentVisibility</a>           | Hides or shows the selected component in the specified configurations in this assembly document.                                                                    |
|    | <a href="#">SetDroppedFileConfigName</a>         | Sets the configuration name for the recently dropped file.                                                                                                          |
|    | <a href="#">SetDroppedFileFeatureName</a>        | Sets the name of the feature for the recently dropped file.                                                                                                         |
|    | <a href="#">SetDroppedFileName</a>               | Sets the new file name for a recently dropped file.                                                                                                                 |
|    | <a href="#">SetIsolateVisibility</a>             | Sets the display characteristics of all of the components not selected to <a href="#">isolate</a> .                                                                 |
|    | <a href="#">SetSpeedPakConfigurations</a>        | Sets the configurations in the selected subassemblies to which to apply SpeedPak in this assembly.                                                                  |
|    | <a href="#">SetSpeedPakToParent</a>              | Switches the selected subassemblies from the SpeedPak configuration to the parent configuration of the SpeedPak configuration.                                      |
|    | <a href="#">ShowExploded</a>                     | <b>Obsolete. Superseded by</b> <a href="#">IAssemblyDoc.ShowExploded2</a>                                                                                           |
|    | <a href="#">ShowExploded2</a>                    | Displays the specified exploded view for the current assembly configuration.                                                                                        |
|    | <a href="#">TemporaryFixGroup</a>                | Temporarily fix or group selected components during such operations as drag, move, rotate, etc.                                                                     |
|    | <a href="#">TemporaryFixGroupExit</a>            | Changes components that were temporarily fixed or grouped back to floating or ungrouped after such operations as drag, move, rotate, etc.                           |
|    | <a href="#">ToolsCheckInterference</a>           | <b>Obsolete. Superseded by</b> <a href="#">IAssemblyDoc.ToolsCheckInterference2</a>                                                                                 |
|    | <a href="#">ToolsCheckInterference2</a>          | Checks for interference between parts in this assembly.                                                                                                             |
|    | <a href="#">TranslateComponent</a>               | Displays the Move Component PropertyManager page.                                                                                                                   |
|    | <a href="#">UnfixComponent</a>                   | Floats the selected component; i.e., makes it movable.                                                                                                              |
|    | <a href="#">UngroupComponents</a>                | Ungroups the grouped components in the selected folder in the FeatureManager design tree.                                                                           |
|    | <a href="#">UpdateBox</a>                        | Updates the bounding box for this assembly.                                                                                                                         |
|    | <a href="#">UpdateFeatureScope</a>               | Updates the feature scope and rebuilds the currently selected assembly feature.                                                                                     |
|   | <a href="#">UpdateSpeedPak</a>                   | Updates out-of-date SpeedPak configurations for the selected subassemblies.                                                                                         |
|  | <a href="#">UpdateToolboxComponent</a>           | Updates SOLIDWORKS Toolbox components in the specified assembly level using the current information in Toolbox settings.                                            |
|  | <a href="#">UseSpeedPak</a>                      | Sets whether to switch the selected subassemblies whose active configuration is the parent configuration of a SpeedPak configuration to the SpeedPak configuration. |
|  | <a href="#">ViewCollapseAssembly</a>             | Collapses the selected exploded view on the Configuration tab of the FeatureManager design tree.                                                                    |
|  | <a href="#">ViewExplodeAssembly</a>              | Explodes the selected exploded view on the Configuration tab of the FeatureManager design tree.                                                                     |
|  | <a href="#">ViewFeatureManagerByDependencies</a> | <b>Obsolete. Superseded by</b> <a href="#">IFeatureManager.ViewDependencies</a>                                                                                     |
|  | <a href="#">ViewFeatureManagerByFeatures</a>     | <b>Obsolete. Superseded by</b> <a href="#">IFeatureManager.ViewFeatures</a>                                                                                         |
|  | <a href="#">ViewFeatureManagerFeatureDetail</a>  | <b>Obsolete. Superseded by</b> <a href="#">IFeatureManager.ShowFeatureDetails</a>                                                                                   |

[Top](#)

## See Also

[IAssemblyDoc Interface](#)  
[SolidWorks.Interop.sldworks Namespace](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

## IAssemblyDoc Interface Members

[See Also](#) [Properties](#) [Methods](#)



[SolidWorks.Interop.sldworks Namespace](#) : IAssemblyDoc Interface

The following tables list the members exposed by [IAssemblyDoc](#).

## Public Properties

| Name                                                                                                                             | Description                                                  |
|----------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------|
|  <a href="#">ClearanceVerificationManager</a> | Gets the clearance verification manager.                     |
|  <a href="#">EnableAssemblyRebuild</a>        | Gets or sets whether to suspend the rebuild of the assembly. |
|  <a href="#">EnablePresentation</a>           | Gets or sets whether the assembly is in presentation mode.   |

|                                                                                   |                                                           |                                                                                                                                                                                                                     |
|-----------------------------------------------------------------------------------|-----------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|   | <a href="#">InterferenceDetectionManager</a>              | Gets the Interference Detection manager, which allows you to run interference detection on an assembly to determine whether components interfere with each other.                                                   |
|  | <a href="#">IsSupportedMatesAvailable</a>                 | Gets whether this assembly contains mates that are supportive of a mate controller.                                                                                                                                 |
|  | <a href="#">LargeDesignReviewTransparencyLevel</a>        | Gets or sets the transparency level of unmodified components in the graphics area of this assembly opened in Large Design Review mode.                                                                              |
|  | <a href="#">LargeDesignReviewTransparencyLevelDynamic</a> | Gets or sets whether to dynamically modify the transparency level of unmodified components in the graphics area when the transparency level slider is moved on the Filter Modified Components PropertyManager page. |
|  | <a href="#">LargeDesignReviewTransparencyLevelEnabled</a> | Gets or sets whether transparency levels are activated for unmodified components in the graphics area for this assembly opened in Large Design Review mode.                                                         |

[Top](#)

## Public Methods

| Name                                                                                | Description                                                                                                                                                                                                                      |
|-------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|    | <a href="#">ActivateGroundPlane</a><br>Activates the ground plane for the specified configurations.                                                                                                                              |
|    | <a href="#">AddComponent</a><br><b>Obsolete. Superseded by <a href="#">IAssemblyDoc::AddComponent4</a>.</b>                                                                                                                      |
|    | <a href="#">AddComponent2</a><br><b>Obsolete. Superseded by <a href="#">IAssemblyDoc::AddComponent4</a>.</b>                                                                                                                     |
|    | <a href="#">AddComponent4</a><br><b>Obsolete. Superseded by <a href="#">IAssemblyDoc::AddComponent5</a>.</b>                                                                                                                     |
|    | <a href="#">AddComponent5</a><br>Adds the specified component for the specified configuration options to this assembly.                                                                                                          |
|    | <a href="#">AddComponentConfiguration</a><br>Adds a new configuration for the last selected assembly component.                                                                                                                  |
|    | <a href="#">AddComponents</a><br><b>Obsolete. Superseded by <a href="#">IAssemblyDoc::AddComponents3</a>.</b>                                                                                                                    |
|    | <a href="#">AddComponents3</a><br>Adds multiple components to the assembly.                                                                                                                                                      |
|    | <a href="#">AddConcentricMateWithTolerance</a><br>Adds a misaligned concentric mate to this assembly.                                                                                                                            |
|    | <a href="#">AddDistanceMate</a><br>Adds a distance mate to this assembly.                                                                                                                                                        |
|   | <a href="#">AddMate</a><br><b>Obsolete. Superseded by <a href="#">IAssemblyDoc::AddMate3</a>.</b>                                                                                                                                |
|  | <a href="#">AddMate2</a><br><b>Obsolete. Superseded by <a href="#">IAssemblyDoc::AddMate3</a>.</b>                                                                                                                               |
|  | <a href="#">AddMate3</a><br><b>Obsolete. Superseded by <a href="#">IAssemblyDoc::AddMate4</a>.</b>                                                                                                                               |
|  | <a href="#">AddMate4</a><br><b>Obsolete. Superseded by <a href="#">IAssemblyDoc::AddMate5</a>.</b>                                                                                                                               |
|  | <a href="#">AddMate5</a><br><b>Obsolete. Superseded by <a href="#">IAssemblyDoc::CreateMate</a>.</b>                                                                                                                             |
|  | <a href="#">AddPipePenetration</a><br>Penetrates the adjacent fitting or pipe with the pipe that ends at the selected sketch point.                                                                                              |
|  | <a href="#">AddPipingFitting</a><br>Adds a pipe fitting to the current piping assembly.                                                                                                                                          |
|  | <a href="#">AddPLMComponent</a><br>Adds the specified component from a 3DEXPERIENCE collaborative space to the specified location in this assembly in <a href="#">SOLIDWORKS Design connected to the 3DEXPERIENCE platform</a> . |
|  | <a href="#">AddSmartComponent</a><br>Adds the specified component at the specified coordinates as a Smart Component to this assembly.                                                                                            |
|  | <a href="#">AddToFeatureScope</a><br>Adds a component to the scope of the currently selected assembly feature.                                                                                                                   |
|  | <a href="#">AssemblyPartToggle</a><br><b>Obsolete. Superseded by <a href="#">IAssemblyDoc::EditAssembly</a> and <a href="#">IAssemblyDoc::EditPart2</a>.</b>                                                                     |
|  | <a href="#">AutoAngleAxis</a><br>Automatically detect the axis for an angle mate.                                                                                                                                                |
|  | <a href="#">AutoExplode</a><br>Automatically generates an exploded view of the current assembly configuration.                                                                                                                   |
|  | <a href="#">AutoMateRepair</a><br>Automatically repairs broken mates in this assembly.                                                                                                                                           |
|  | <a href="#">CollectAllSupportiveMates</a><br>Gets all mates supportive of a mate controller in this assembly.                                                                                                                    |
|  | <a href="#">CompConfigProperties</a><br><b>Obsolete. Superseded by <a href="#">IAssemblyDoc::CompConfigProperties4</a>.</b>                                                                                                      |
|  | <a href="#">CompConfigProperties2</a><br><b>Obsolete. Superseded by <a href="#">IAssemblyDoc::CompConfigProperties4</a>.</b>                                                                                                     |
|  | <a href="#">CompConfigProperties3</a><br><b>Obsolete. Superseded by <a href="#">IAssemblyDoc::CompConfigProperties4</a>.</b>                                                                                                     |
|  | <a href="#">CompConfigProperties4</a><br><b>Obsolete. Superseded by <a href="#">IAssemblyDoc::CompConfigProperties5</a>.</b>                                                                                                     |
|  | <a href="#">CompConfigProperties5</a><br><b>Obsolete. Superseded by <a href="#">IAssemblyDoc::CompConfigProperties6</a>.</b>                                                                                                     |
|  | <a href="#">CompConfigProperties6</a><br>Sets the properties for the selected <a href="#">component</a> in the specified <a href="#">configuration</a> .                                                                         |
|  | <a href="#">ComponentReload</a><br><b>Obsolete. Superseded by <a href="#">IAssemblyDoc::ComponentReload2</a>.</b>                                                                                                                |
|  | <a href="#">ComponentReload2</a><br>Reloads and/or sets the read-only state of the specified component.                                                                                                                          |
|  | <a href="#">CopyWithMates</a><br><b>Obsolete. Superseded by <a href="#">IAssemblyDoc::CopyWithMates2</a>.</b>                                                                                                                    |

|                                                                                     |                                                  |                                                                                                                      |
|-------------------------------------------------------------------------------------|--------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
|    | <a href="#">CopyWithMates2</a>                   | Copies one or more components with mates in this assembly.                                                           |
|    | <a href="#">CreateExplodedView</a>               | Creates an explode view of the active assembly configuration.                                                        |
|    | <a href="#">CreateMate</a>                       | Creates a mate with the specified feature data object.                                                               |
|    | <a href="#">CreateMateData</a>                   | Creates a mate feature data object for the specified mate type.                                                      |
|    | <a href="#">CreateSmartComponent</a>             | Creates a Smart Component.                                                                                           |
|    | <a href="#">CreateSpeedPak</a>                   | Creates the specified type of SpeedPak for the active configurations of the selected subassemblies in this assembly. |
|    | <a href="#">DeleteSelections</a>                 | Delete either the selected components of a subassembly or the subassembly of the selected component.                 |
|    | <a href="#">DissolveComponentPattern</a>         | Dissolves the selected component patterns.                                                                           |
|    | <a href="#">DissolveSubAssembly</a>              | Dissolves the selected subassembly in this assembly.                                                                 |
|    | <a href="#">EditAssembly</a>                     | Switches back to the assembly document for editing.                                                                  |
|    | <a href="#">EditConcentricMate</a>               | Edits a misaligned concentric mate.                                                                                  |
|    | <a href="#">EditDistanceMate</a>                 | Edits a distance mate.                                                                                               |
|    | <a href="#">EditMate</a>                         | Obsolete. Superseded by <a href="#">IAssemblyDoc::EditMate2</a> .                                                    |
|    | <a href="#">EditMate2</a>                        | Obsolete. Superseded by <a href="#">IAssemblyDoc::EditMate3</a> .                                                    |
|    | <a href="#">EditMate3</a>                        | Obsolete. Superseded by <a href="#">IAssemblyDoc::EditMate4</a> .                                                    |
|    | <a href="#">EditMate4</a>                        | Obsolete. See the Remarks in each standard, advanced, and mechanical mate's feature data interface.                  |
|    | <a href="#">EditPart</a>                         | Obsolete. Superseded by <a href="#">IAssemblyDoc::EditPart2</a> .                                                    |
|    | <a href="#">EditPart2</a>                        | Edits the selected part within the context of an assembly.                                                           |
|    | <a href="#">EditRebuild</a>                      | Obsolete. Superseded by <a href="#">IModelDoc2::EditRebuild3</a> .                                                   |
|    | <a href="#">ExitIsolate</a>                      | Exits <a href="#">isolating the selected components</a> and returns the assembly to its original display state.      |
|    | <a href="#">FeatureByName</a>                    | Returns the <a href="#">IFeature</a> object for the named feature in the assembly.                                   |
|   | <a href="#">FeatureExtrusion</a>                 | Obsolete. Superseded by <a href="#">IFeatureManager::FeatureExtrusion</a> .                                          |
|  | <a href="#">FileDeriveComponentPart</a>          | Creates a new part document from the currently selected assembly component.                                          |
|  | <a href="#">FixComponent</a>                     | Fixes the selected component; i.e., makes it immovable.                                                              |
|  | <a href="#">ForceRebuild</a>                     | Obsolete. Superseded by <a href="#">IModelDoc2::ForceRebuild3</a> .                                                  |
|  | <a href="#">ForceRebuild2</a>                    | Obsolete. Superseded by <a href="#">IModelDoc2::ForceRebuild3</a> .                                                  |
|  | <a href="#">ForceUpdateElectricalData</a>        | Obsolete. Superseded by <a href="#">IAssemblyDoc::ForceUpdateElectricalData2</a> .                                   |
|  | <a href="#">ForceUpdateElectricalData2</a>       | Forces an update of electrical data.                                                                                 |
|  | <a href="#">GetActiveGroundPlane</a>             | Gets the active ground plane for the specified configurations.                                                       |
|  | <a href="#">GetAdvancedSelection</a>             | Gets the advanced component selection criteria object for this assembly.                                             |
|  | <a href="#">GetBox</a>                           | Gets the bounding box.                                                                                               |
|  | <a href="#">GetComponentById</a>                 | Gets a top-level assembly component using its component ID.                                                          |
|  | <a href="#">GetComponentByName</a>               | Gets the specified top-level assembly component.                                                                     |
|  | <a href="#">GetComponentCount</a>                | Gets the number of components in the active configuration of this assembly.                                          |
|  | <a href="#">GetComponents</a>                    | Gets all of the components in the active configuration of this assembly.                                             |
|  | <a href="#">GetDragOperator</a>                  | Gets the drag operator for dynamic drag operations in this assembly.                                                 |
|  | <a href="#">GetDroppedAtEntity</a>               | Gets a pointer to the entity where a file is dropped into this assembly.                                             |
|  | <a href="#">GetEditTarget</a>                    | Gets the model document that is currently being edited.                                                              |
|  | <a href="#">GetEditTargetComponent</a>           | Gets the component that is currently being edited.                                                                   |
|  | <a href="#">GetExplodedViewConfigurationName</a> | Gets the name of the configuration for the specified exploded view.                                                  |
|  | <a href="#">GetExplodedViewCount</a>             | Obsolete. Superseded by <a href="#">IAssemblyDoc::GetExplodedViewCount2</a> .                                        |
|  | <a href="#">GetExplodedViewCount2</a>            | Gets the number of exploded views in the specified configuration.                                                    |
|  | <a href="#">GetExplodedViewNames</a>             | Obsolete. Superseded by <a href="#">IAssemblyDoc::GetExplodedViewNames2</a> .                                        |

|                                                                                     |                                                 |                                                                                                                                |
|-------------------------------------------------------------------------------------|-------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
|     |                                                 |                                                                                                                                |
|    | <a href="#">GetExplodedViewNames2</a>           | Gets the names of the exploded views in the specified configuration.                                                           |
|    | <a href="#">GetFeatureScope</a>                 | Gets the components affected by this feature.                                                                                  |
|    | <a href="#">GetFeatureScopeCount</a>            | Gets the number of components affected by this feature.                                                                        |
|    | <a href="#">GetFirstMember</a>                  | Obsolete. Superseded by <a href="#">IComponent2::GetChildren</a> .                                                             |
|    | <a href="#">GetLargeDesignReviewState</a>       | Gets the Large Design Review state of this assembly.                                                                           |
|    | <a href="#">GetLightWeightComponentCount</a>    | Gets the number of lightweight components in the assembly.                                                                     |
|    | <a href="#">GetPhysicalSimulationComponents</a> | Obsolete. Not superseded.                                                                                                      |
|    | <a href="#">GetRouteManager</a>                 | Gets the SOLIDWORKS Routing API.                                                                                               |
|    | <a href="#">GetSelectedMember</a>               | Obsolete. Superseded by <a href="#">ISelectionMgr::GetSelectedObject6</a> .                                                    |
|    | <a href="#">GetSimulation</a>                   | Obsolete. Not superseded.                                                                                                      |
|    | <a href="#">GetUnloadedComponentNames</a>       | Gets the unloaded components' paths, referenced configuration names, reasons why they are unloaded, document types, and names. |
|    | <a href="#">GetVisibleComponentsInView</a>      | Gets a list of visible components in this assembly to save as solid bodies.                                                    |
|    | <a href="#">GetVisibleComponentsInViewCount</a> | Gets the number of visible components in this assembly.                                                                        |
|    | <a href="#">HasUnloadedComponents</a>           | Gets whether this assembly has hidden or suppressed unloaded components.                                                       |
|    | <a href="#">HideComponent</a>                   | Obsolete. Superseded by <a href="#">IModelDoc2::HideComponent2</a> .                                                           |
|    | <a href="#">IAddComponent2</a>                  | Obsolete. Superseded by <a href="#">IAssemblyDoc::AddComponent4</a> .                                                          |
|    | <a href="#">IAddComponent3</a>                  | Obsolete. Superseded by <a href="#">IAssemblyDoc::AddComponent4</a> .                                                          |
|    | <a href="#">IAddComponents</a>                  | Obsolete. Superseded by <a href="#">IAssemblyDoc::IAddComponents2</a> .                                                        |
|    | <a href="#">IAddComponents2</a>                 | Obsolete. Superseded by <a href="#">IAssemblyDoc::IAddComponents3</a> .                                                        |
|    | <a href="#">IAddComponents3</a>                 | Adds multiple components to the assembly.                                                                                      |
|   | <a href="#">IComponentReload2</a>               | Obsolete. Superseded by <a href="#">IAssemblyDoc::ReplaceComponents</a> .                                                      |
|  | <a href="#">IComponentReload3</a>               | Obsolete. Superseded by <a href="#">IAssemblyDoc::ReplaceComponents</a> .                                                      |
|  | <a href="#">IFeatureByName</a>                  | Returns the <a href="#">IFeature</a> object for the named feature in the assembly.                                             |
|  | <a href="#">IGetBox</a>                         | Gets the bounding box.                                                                                                         |
|  | <a href="#">IGetComponents</a>                  | Gets all of the components in the active configuration of this assembly.                                                       |
|  | <a href="#">IGetDragOperator</a>                | Gets the drag operator for dynamic drag operations in this assembly.                                                           |
|  | <a href="#">IGetEditTarget</a>                  | Obsolete. Superseded by <a href="#">IAssemblyDoc::IGetEditTarget2</a> .                                                        |
|  | <a href="#">IGetEditTarget2</a>                 | Gets the model document that is currently being edited.                                                                        |
|  | <a href="#">IGetExplodedViewNames</a>           | Obsolete. Superseded by <a href="#">IAssemblyDoc::GetExplodedViewNames2</a> .                                                  |
|  | <a href="#">IGetFeatureScope</a>                | Gets the components affected by this feature.                                                                                  |
|  | <a href="#">IGetFirstMember</a>                 | Obsolete. Superseded by <a href="#">IComponent2::GetChildren</a> .                                                             |
|  | <a href="#">IGetSelectedMember</a>              | Obsolete. Superseded by <a href="#">ISelectionMgr::GetSelectedObject6</a> .                                                    |
|  | <a href="#">IGetVisibleComponentsInView</a>     | Gets a list of visible components in this assembly to save as solid bodies.                                                    |
|  | <a href="#">IMirrorComponents</a>               | Obsolete. Superseded by <a href="#">IAssemblyDoc::MirrorComponents2</a> .                                                      |
|  | <a href="#">InsertCavity</a>                    | Obsolete. Superseded by <a href="#">IAssemblyDoc::InsertCavity4</a> .                                                          |
|  | <a href="#">InsertCavity2</a>                   | Obsolete. Superseded by <a href="#">IAssemblyDoc::InsertCavity4</a> .                                                          |
|  | <a href="#">InsertCavity3</a>                   | Obsolete. Superseded by <a href="#">IAssemblyDoc::InsertCavity4</a> .                                                          |
|  | <a href="#">InsertCavity4</a>                   | Inserts a cavity to the active part using a selected component.                                                                |
|  | <a href="#">InsertDerivedPattern</a>            | Obsolete. See <a href="#">IFeatureManager::CreateFeature</a> and the Remarks in <a href="#">IDerivedPatternFeatureData</a> .   |
|  | <a href="#">InsertEnvelope</a>                  | Adds an envelope in the specified configuration name in this assembly.                                                         |
|  | <a href="#">InsertImportedComponent</a>         | Inserts a third-party native CAD part or assembly into the current configuration of this assembly.                             |

|                                                                                     |                                                       |                                                                                                                                                                                               |
|-------------------------------------------------------------------------------------|-------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|    | <a href="#">InsertJoin</a>                            | Obsolete. Superseded by <a href="#">IAssemblyDoc::InsertJoin2</a> .                                                                                                                           |
|    | <a href="#">InsertJoin2</a>                           | Constructs a feature from merged selected components.                                                                                                                                         |
|    | <a href="#">InsertLoadReference</a>                   | Creates a mate load reference to the specified or selected mate.                                                                                                                              |
|    | <a href="#">InsertNewAssembly</a>                     | Creates a new virtual sub-assembly and optionally saves it to the specified file.                                                                                                             |
|    | <a href="#">InsertNewPart</a>                         | Obsolete. Superseded by <a href="#">IAssemblyDoc::InsertNewPart2</a> .                                                                                                                        |
|    | <a href="#">InsertNewPart2</a>                        | Inserts a new part on the specified face or plane.                                                                                                                                            |
|    | <a href="#">InsertNewVirtualAssembly</a>              | Creates a new assembly from this assembly and saves it internally as a virtual component.                                                                                                     |
|    | <a href="#">InsertNewVirtualPart</a>                  | Creates a new part in the context of an assembly and saves the part internally in the assembly file as a virtual component.                                                                   |
|    | <a href="#">InsertWeld</a>                            | Obsolete. Do not use. Superseded by <a href="#">IFeatureManager::InsertCosmeticWeldBead</a> .                                                                                                 |
|    | <a href="#">InsertWeld2</a>                           | Obsolete. Do not use. Superseded by <a href="#">IFeatureManager::InsertCosmeticWeldBead</a> .                                                                                                 |
|    | <a href="#">IReorderComponents</a>                    | Moves components to a different location in the FeatureManager tree.                                                                                                                          |
|    | <a href="#">IReorganizeComponents</a>                 | Reorganizes an assembly's structure by moving the selected components to the selected assembly or sub-assembly.                                                                               |
|    | <a href="#">IsComponentTreeValid</a>                  | Checks the validity of the component tree for this assembly.                                                                                                                                  |
|    | <a href="#">ISetComponentState</a>                    | Sets the suppression state for the specified components.                                                                                                                                      |
|    | <a href="#">ISetComponentVisibility</a>               | Hides or shows the selected component in the specified configurations in this assembly document.                                                                                              |
|    | <a href="#">Isolate</a>                               | Isolates the selected components.                                                                                                                                                             |
|    | <a href="#">IsRouteAssembly</a>                       | Gets whether the assembly document is a routing assembly.                                                                                                                                     |
|    | <a href="#">IToolsCheckInterference2</a>              | Obsolete. See <a href="#">IAssemblyDoc::IToolsCheckInterference3</a> .                                                                                                                        |
|    | <a href="#">IToolsCheckInterference3</a>              | Obsolete.                                                                                                                                                                                     |
|    | <a href="#">LightweightAllResolved</a>                | Sets to lightweight all resolved child components of the selected components.                                                                                                                 |
|    | <a href="#">MakeAssemblyFromSelectedComponents</a>    | Creates a new assembly comprised of the selected components of this assembly.                                                                                                                 |
|   | <a href="#">MakeIndependent</a>                       | Makes the selected component independent.                                                                                                                                                     |
|  | <a href="#">MakeLightWeight</a>                       | Sets the selected components to lightweight.                                                                                                                                                  |
|  | <a href="#">MirrorComponents</a>                      | Obsolete. Superseded by <a href="#">IAssemblyDoc::MirrorComponents2</a> .                                                                                                                     |
|  | <a href="#">MirrorComponents2</a>                     | Obsolete. Superseded by <a href="#">IAssemblyDoc::MirrorComponents3</a> .                                                                                                                     |
|  | <a href="#">MirrorComponents3</a>                     | Obsolete. Superseded by <a href="#">IFeatureManager::CreateDefinition</a> , <a href="#">IFeatureManager::CreateFeature</a> , and <a href="#">IMirrorComponentFeatureData</a> .                |
|  | <a href="#">OpenCompFile</a>                          | Obsolete. Superseded by <a href="#">ISldWorks::ActivateDoc2</a> and <a href="#">ISldWorks::OpenDoc6</a> .                                                                                     |
|  | <a href="#">RemoveFromFeatureScope</a>                | Removes a component from the scope of the currently selected assembly feature.                                                                                                                |
|  | <a href="#">ReorderComponents</a>                     | Moves components to a different location in the FeatureManager design tree.                                                                                                                   |
|  | <a href="#">ReorganizeComponents</a>                  | Reorganizes an assembly's structure by moving the selected components to the selected assembly or sub-assembly.                                                                               |
|  | <a href="#">ReplaceComponents</a>                     | Obsolete. Superseded by <a href="#">IAssemblyDoc::ReplaceComponents2</a> .                                                                                                                    |
|  | <a href="#">ReplaceComponents2</a>                    | Replaces one or more selected components with another model.                                                                                                                                  |
|  | <a href="#">ReplacePLMComponents</a>                  | Replaces a selected <a href="#">SOLIDWORKS Design connected to the 3DEXPERIENCE platform</a> component in this assembly with the specified component from a 3DEXPERIENCE collaborative space. |
|  | <a href="#">ResolveAllLightweight</a>                 | Resolves all lightweight child components of the selected components                                                                                                                          |
|  | <a href="#">ResolveAllLightWeightComponents</a>       | Resolves the lightweight components in this assembly.                                                                                                                                         |
|  | <a href="#">ResolveOutOfDateLightWeightComponents</a> | Resolves the selected out-of-date lightweight component, and any out-of-date lightweight sub-components, in the assembly.                                                                     |
|  | <a href="#">RotateComponent</a>                       | Displays the Rotate Component PropertyManager page.                                                                                                                                           |
|  | <a href="#">RotateComponentAxis</a>                   | Rotates the component axis by a fixed amount.                                                                                                                                                 |
|  | <a href="#">SaveIsolate</a>                           | Saves the display characteristics of the <a href="#">isolated components</a> to a new display state.                                                                                          |
|  | <a href="#">SelectComponentsBySize</a>                | Selects assembly components by percent of assembly size.                                                                                                                                      |
|  | <a href="#">SelectiveOpen</a>                         | Selectively opens the components of an assembly that is opened in Large Design Review mode.                                                                                                   |
|  | <a href="#">SetComponentState</a>                     | Sets the suppression state for the specified components.                                                                                                                                      |
|  | <a href="#">SetComponentSuppression</a>               | Suppresses, resolves, or sets to lightweight selected components of this assembly in the active configuration only.                                                                           |

|                                                                                     |                                                  |                                                                                                                                                                     |
|-------------------------------------------------------------------------------------|--------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|     |                                                  |                                                                                                                                                                     |
|    | <a href="#">SetComponentTransparent</a>          | Enables or disables transparency on the selected components.                                                                                                        |
|    | <a href="#">SetComponentVisibility</a>           | Hides or shows the selected component in the specified configurations in this assembly document.                                                                    |
|    | <a href="#">SetDroppedFileConfigName</a>         | Sets the configuration name for the recently dropped file.                                                                                                          |
|    | <a href="#">SetDroppedFileFeatureName</a>        | Sets the name of the feature for the recently dropped file.                                                                                                         |
|    | <a href="#">SetDroppedFileName</a>               | Sets the new file name for a recently dropped file.                                                                                                                 |
|    | <a href="#">SetIsolateVisibility</a>             | Sets the display characteristics of all of the components not selected to <a href="#">isolate</a> .                                                                 |
|    | <a href="#">SetSpeedPakConfigurations</a>        | Sets the configurations in the selected subassemblies to which to apply SpeedPak in this assembly.                                                                  |
|    | <a href="#">SetSpeedPakToParent</a>              | Switches the selected subassemblies from the SpeedPak configuration to the parent configuration of the SpeedPak configuration.                                      |
|    | <a href="#">ShowExploded</a>                     | Obsolete. Superseded by <a href="#">IAssemblyDoc::ShowExploded2</a> .                                                                                               |
|    | <a href="#">ShowExploded2</a>                    | Displays the specified exploded view for the current assembly configuration.                                                                                        |
|    | <a href="#">TemporaryFixGroup</a>                | Temporarily fix or group selected components during such operations as drag, move, rotate, etc.                                                                     |
|    | <a href="#">TemporaryFixGroupExit</a>            | Changes components that were temporarily fixed or grouped back to floating or ungrouped after such operations as drag, move, rotate, etc.                           |
|    | <a href="#">ToolsCheckInterference</a>           | Obsolete. Superseded by <a href="#">IAssemblyDoc::ToolsCheckInterference2</a> .                                                                                     |
|    | <a href="#">ToolsCheckInterference2</a>          | Checks for interference between parts in this assembly.                                                                                                             |
|    | <a href="#">TranslateComponent</a>               | Displays the Move Component PropertyManager page.                                                                                                                   |
|    | <a href="#">UnfixComponent</a>                   | Floats the selected component; i.e., makes it movable.                                                                                                              |
|    | <a href="#">UngroupComponents</a>                | Ungroups the grouped components in the selected folder in the FeatureManager design tree.                                                                           |
|    | <a href="#">UpdateBox</a>                        | Updates the bounding box for this assembly.                                                                                                                         |
|    | <a href="#">UpdateFeatureScope</a>               | Updates the feature scope and rebuilds the currently selected assembly feature.                                                                                     |
|   | <a href="#">UpdateSpeedPak</a>                   | Updates out-of-date SpeedPak configurations for the selected subassemblies.                                                                                         |
|  | <a href="#">UpdateToolboxComponent</a>           | Updates SOLIDWORKS Toolbox components in the specified assembly level using the current information in Toolbox settings.                                            |
|  | <a href="#">UseSpeedPak</a>                      | Sets whether to switch the selected subassemblies whose active configuration is the parent configuration of a SpeedPak configuration to the SpeedPak configuration. |
|  | <a href="#">ViewCollapseAssembly</a>             | Collapses the selected exploded view on the Configuration tab of the FeatureManager design tree.                                                                    |
|  | <a href="#">ViewExplodeAssembly</a>              | Explodes the selected exploded view on the Configuration tab of the FeatureManager design tree.                                                                     |
|  | <a href="#">ViewFeatureManagerByDependencies</a> | Obsolete. Superseded by <a href="#">IFeatureManager::ViewDependencies</a> .                                                                                         |
|  | <a href="#">ViewFeatureManagerByFeatures</a>     | Obsolete. Superseded by <a href="#">IFeatureManager::ViewFeatures</a> .                                                                                             |
|  | <a href="#">ViewFeatureManagerFeatureDetail</a>  | Obsolete. Superseded by <a href="#">IFeatureManager::ShowFeatureDetails</a> .                                                                                       |

[Top](#)

## See Also

[IAssemblyDoc Interface](#)  
[SolidWorks.Interop.sldworks Namespace](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

## IAssemblyDoc Interface Properties

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) : IAssemblyDoc Interface

For a list of all members of this type, see [IAssemblyDoc members](#).

## Public Properties

| Name                                                                                                                             | Description                                                  |
|----------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------|
|  <a href="#">ClearanceVerificationManager</a> | Gets the clearance verification manager.                     |
|  <a href="#">EnableAssemblyRebuild</a>        | Gets or sets whether to suspend the rebuild of the assembly. |
|  <a href="#">EnablePresentation</a>           | Gets or sets whether the assembly is in presentation mode.   |

|                                                                                                                                             |                                                                                                                                                                                                                     |
|---------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  <a href="#">InterferenceDetectionManager</a>               | Gets the Interference Detection manager, which allows you to run interference detection on an assembly to determine whether components interfere with each other.                                                   |
|  <a href="#">IsSupportedMatesAvailable</a>                 | Gets whether this assembly contains mates that are supportive of a mate controller.                                                                                                                                 |
|  <a href="#">LargeDesignReviewTransparencyLevel</a>        | Gets or sets the transparency level of unmodified components in the graphics area of this assembly opened in Large Design Review mode.                                                                              |
|  <a href="#">LargeDesignReviewTransparencyLevelDynamic</a> | Gets or sets whether to dynamically modify the transparency level of unmodified components in the graphics area when the transparency level slider is moved on the Filter Modified Components PropertyManager page. |
|  <a href="#">LargeDesignReviewTransparencyLevelEnabled</a> | Gets or sets whether transparency levels are activated for unmodified components in the graphics area for this assembly opened in Large Design Review mode.                                                         |

[Top](#)

## See Also

[IAssemblyDoc Interface](#)  
[SolidWorks.Interop.sldworks Namespace](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

## ActivateGroundPlane Method (IAssemblyDoc)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IAssemblyDoc Interface](#) : ActivateGroundPlane Method (IAssemblyDoc)

Activates the ground plane for the specified configurations.

## .NET Syntax

### Visual Basic (Declaration)

```
Function ActivateGroundPlane( _
    ByVal Config\_opt As System.Integer, _
    ByVal Config\_names As System.Object _
) As System.Boolean
```

### Visual Basic (Usage)



```
Dim instance As IAssemblyDoc
Dim Config_opt As System.Integer
Dim Config_names As System.Object
Dim value As System.Boolean

value = instance.ActivateGroundPlane(Config_opt, Config_names)
```

### C#

```
System.bool ActivateGroundPlane(
    System.int Config\_opt,
    System.object Config\_names
)
```

### C++/CLI

```
System.bool ActivateGroundPlane(
    & System.int Config\_opt,
    & System.Object^ Config\_names
)
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

### Parameters

*Config\_opt*

Configurations in which to activate a ground plane as defined in [swInConfigurationOpts\\_e](#):

- swThisConfiguration
- swAllConfigurations
- swSpecifyConfiguration

*Config\_names*

Array of configurations in which to activate a ground plane; valid only if Config\_opt is set to swInConfiguration\_e.swSpecifyConfiguration (see **Remarks**)

### Return Value

True if ground plane successfully activated in the specified configurations, false if not

## Visual Basic for Applications (VBA) Syntax

See [AssemblyDoc::ActivateGroundPlane](#).

## Example

[Insert and Activate Ground Plane \(VBA\)](#)

## Remarks

To populate config\_names, use [IModelDoc2::GetConfigurationNames](#).

Before calling this method, use [IModelDocExtension::SelectByID2](#) to select the ground plane to activate in the specified configurations.

Only one ground plane can be active at a given time in each assembly configuration.

## See Also

[IAssemblyDoc Interface](#)  
[IAssemblyDoc Members](#)  
[IAssemblyDoc::GetActiveGroundPlane Method \(\)](#)  
[IFeatureManager::InsertGroundPlane Method \(\)](#)

## Availability

SOLIDWORKS 2018 FCS, Revision Number 26.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

## AddComponent Method (IAssemblyDoc)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IAssemblyDoc Interface](#) : AddComponent Method (IAssemblyDoc)

Obsolete. Superseded by [IAssemblyDoc::AddComponent4](#).

## .NET Syntax

Visual Basic (Declaration)

```
Function AddComponent( _  
    ByVal CompName As System.String, _  
    ByVal X As System.Double, _  
    ByVal Y As System.Double, _  
    ByVal Z As System.Double _  
    ) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IAssemblyDoc  
Dim CompName As System.String  
Dim X As System.Double  
Dim Y As System.Double  
Dim Z As System.Double  
Dim value As System.Boolean  
  
value = instance.AddComponent(CompName, X, Y, Z)
```

C#

```
System.bool AddComponent(  
    System.string CompName,  
    System.double X,  
    System.double Y,  
    System.double Z  
)
```

C++/CLI

```
System.bool AddComponent(  
& System.String^ CompName,  
& System.double X,
```

```
& System.double Y,  
& System.double Z  
)
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Parameters

*CompName*

*X*

*Y*

*Z*

### Visual Basic for Applications (VBA) Syntax

See [AssemblyDoc::AddComponent](#).

### See Also

[IAssemblyDoc Interface](#)  
[IAssemblyDoc Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

## AddComponent2 Method (IAssemblyDoc)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IAssemblyDoc Interface](#) : AddComponent2 Method (IAssemblyDoc)

Obsolete. Superseded by [IAssemblyDoc::AddComponent4](#).

### .NET Syntax

#### Visual Basic (Declaration)

```
Function AddComponent2( _  
    ByVal CompName As System.String, _  
    ByVal X As System.Double, _  
    ByVal Y As System.Double, _  
    ByVal Z As System.Double _  
) As System.Object
```

#### Visual Basic (Usage)

```
Dim instance As IAssemblyDoc  
Dim CompName As System.String  
Dim X As System.Double  
Dim Y As System.Double  
Dim Z As System.Double  
Dim value As System.Object
```

```
value = instance.AddComponent2(CompName, X, Y, Z)
```

#### C#

```
System.object AddComponent2(  
    System.string CompName,  
    System.double X,  
    System.double Y,  
    System.double Z  
)
```

#### C++/CLI

```
System.Object^ AddComponent2(  
& System.String^ CompName,  
& System.double X,  
& System.double Y,  
& System.double Z  
)
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Parameters

*CompName*

*X*

*Y*

*Z*

### Visual Basic for Applications (VBA) Syntax

See [AssemblyDoc::AddComponent2](#).

### See Also

[IAssemblyDoc Interface](#)  
[IAssemblyDoc Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

## AddComponent4 Method (IAssemblyDoc)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IAssemblyDoc Interface](#) : AddComponent4 Method (IAssemblyDoc)

Obsolete. Superseded by [IAssemblyDoc::AddComponent5](#).

### .NET Syntax

#### Visual Basic (Declaration)

```
Function AddComponent4( _  
    ByVal CompName As System.String, _  
    ByVal ConfigName As System.String, _  
    ByVal X As System.Double, _  
    ByVal Y As System.Double, _  
    ByVal Z As System.Double _  
) As Component2
```

#### Visual Basic (Usage)

```
Dim instance As IAssemblyDoc  
Dim CompName As System.String  
Dim ConfigName As System.String  
Dim X As System.Double  
Dim Y As System.Double  
Dim Z As System.Double  
Dim value As Component2
```

```
value = instance.AddComponent4(CompName, ConfigName, X, Y, Z)
```

#### C#

```
Component2 AddComponent4(  
    System.string CompName,  
    System.string ConfigName,  
    System.double X,  
    System.double Y,  
    System.double Z  
)
```

#### C++/CLI



```

Component2^ AddComponent4(
& System.String^ CompName,
& System.String^ ConfigName,
& System.double X,
& System.double Y,
& System.double Z
)

```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Parameters

*CompName*

Path name of a loaded part or assembly to add as a component (see **Remarks**)

*ConfigName*

Name of the configuration from which to load the component (see **Remarks**)

*X*

X coordinate of the component center

*Y*

Y coordinate of the component center

*Z*

Z coordinate of the component center

#### Return Value

[IComponent2](#)

### Visual Basic for Applications (VBA) Syntax

See [AssemblyDoc::AddComponent4](#).

### Example

[Add Component and Mate \(C++\)](#)

### Remarks

The specified file must be loaded in memory. A file is loaded into memory when you load the file in your SOLIDWORKS Design session ([ISldWorks::OpenDoc6](#)) or open an assembly that already contains the file.

If configName is empty or specifies a configuration that does not exist, then the current configuration is used.

If you want to add the component at a position relative to the root component, use [IComponent2::Transform2](#) to reposition the component immediately after calling this method. Or, you can use [IAssemblyDoc::AddComponents](#) or [IAssemblyDoc::AddComponents2](#) to specify the transform when inserting a component, insert as many components as you want at once, and rotate and scale the component through the transform matrix.

**IMPORTANT:** The x, y, and z parameters of this method are relative to the bounding box of the component. Only use this method if you want to approximately position the component. Use [IComponent2::Transform2](#) if you want to more precisely position the component.

### See Also

[IAssemblyDoc Interface](#)

[IAssemblyDoc Members](#)

[IAssemblyDoc::InsertEnvelope Method](#)

[IAssemblyDoc::ReplaceComponents Method](#)

### Availability

SOLIDWORKS 2003 SP5, Revision Number 11.5

SOLIDWORKS API Help

[Send comments on this topic.](#)

## AddComponent5 Method (IAssemblyDoc)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IAssemblyDoc Interface](#) : AddComponent5 Method (IAssemblyDoc)

Adds the specified component for the specified configuration options to this assembly.

### .NET Syntax

Visual Basic (Declaration)

```

Function AddComponent5( _
    ByVal CompName As System.String, _
    ByVal ConfigOption As System.Integer, _
    ByVal NewConfigName As System.String, _
    ByVal UseConfigForPartReferences As System.Boolean, _
    ByVal ExistingConfigName As System.String, _
    ByVal X As System.Double, _
    ByVal Y As System.Double, _
    ByVal Z As System.Double _
) As Component2

```

Visual Basic (Usage)



```

Dim instance As IAssemblyDoc
Dim CompName As System.String
Dim ConfigOption As System.Integer
Dim NewConfigName As System.String
Dim UseConfigForPartReferences As System.Boolean
Dim ExistingConfigName As System.String
Dim X As System.Double
Dim Y As System.Double
Dim Z As System.Double
Dim value As Component2

```

```

value = instance.AddComponent5(CompName, ConfigOption, NewConfigName, UseConfigForPartReferences,
ExistingConfigName, X, Y, Z)

```

C#

```

Component2 AddComponent5(
    System.string CompName,
    System.int ConfigOption,
    System.string NewConfigName,
    System.bool UseConfigForPartReferences,
    System.string ExistingConfigName,
    System.double X,
    System.double Y,
    System.double Z
)

```

C++/CLI

```

Component2^ AddComponent5(
    & System.String^ CompName,
    & System.int ConfigOption,
    & System.String^ NewConfigName,
    & System.bool UseConfigForPartReferences,
    & System.String^ ExistingConfigName,
    & System.double X,
    & System.double Y,
    & System.double Z
)

```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Parameters

*CompName*

Path name of a pre-loaded part or assembly to add as a component (see **Remarks**)

*ConfigOption*

Mode in which to open the document as specified in [swAddComponentConfigOptions\\_e](#)

*NewConfigName*

Name for the new assembly configuration;

valid only if ConfigOption =

- [swAddComponentConfigOptions\\_e.swAddComponentConfigOptions\\_NewConfigWithAllReferenceModels](#) - or -
- [swAddComponentConfigOptions\\_e.swAddComponentConfigOptions\\_NewConfigWithAsmStructure](#)

*UseConfigForPartReferences*

If true, the configuration specified in ExistingConfigName is used

*ExistingConfigName*

Name of the configuration of the loaded component; valid only if UseConfigForPartReferences = true

*X*

X coordinate of the component center

*Y*

Y coordinate of the component center  
Z  
Z coordinate of the component center

#### Return Value

[IComponent2](#)

### Visual Basic for Applications (VBA) Syntax

See [AssemblyDoc::AddComponent5](#).

### Example

[Insert New Instance of Virtual Component \(VBA\)](#)  
[Insert New Instance of Virtual Component \(VB.NET\)](#)  
[Insert New Instance of Virtual Component \(C#\)](#)  
[Add Component and Mate \(VBA\)](#)  
[Add Component and Mate \(VB.NET\)](#)  
[Add Component and Mate \(C#\)](#)

### Remarks

The specified file must be loaded in memory. A file is loaded into memory when you load the file in your SOLIDWORKS Design session ([ISldWorks::OpenDoc6](#)) or open an assembly that already contains the file.

To add a new instance of an existing virtual component, use [IComponent2::GetPathName](#) to get the virtual component's path and file name.

If you want to add the component at a position relative to the root component, use [IComponent2::Transform2](#) to reposition the component immediately after calling this method. Or, you can use [IAssemblyDoc::AddComponents](#) or [IAssemblyDoc::AddComponents2](#) to specify the transform when inserting a component, insert as many components as you want at once, and rotate and scale the component through the transform matrix.

**IMPORTANT:** X, Y, and Z are relative to the bounding box of the component. Use this method to approximately position the component. Then use [IComponent2::Transform2](#) to more precisely position the component.

### See Also

[IAssemblyDoc Interface](#)  
[IAssemblyDoc Members](#)  
[IAssemblyDoc::InsertEnvelope Method](#)  
[IAssemblyDoc::ReplaceComponents Method](#)  
[IAssemblyDoc::CopyWithMates Method](#)  
[IAssemblyDoc::InsertImportedComponent Method](#)

### Availability

SOLIDWORKS 2010 FCS, Revision Number 18.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

## AddComponentConfiguration Method (IAssemblyDoc)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IAssemblyDoc Interface](#) : AddComponentConfiguration Method (IAssemblyDoc)

Adds a new configuration for the last selected assembly component.

### .NET Syntax

Visual Basic (Declaration)

```
Function AddComponentConfiguration( _  
    ByVal Name As System.String, _  
    ByVal Comment As System.String, _  
    ByVal AlternateName As System.String, _  
    ByVal Options As System.Integer _  
    ) As Configuration
```

Visual Basic (Usage)

```
Dim instance As IAssemblyDoc  
Dim Name As System.String  
Dim Comment As System.String
```



```
Dim AlternateName As System.String
Dim Options As System.Integer
Dim value As Configuration

value = instance.AddComponentConfiguration(Name, Comment, AlternateName, Options)
```

C#

```
Configuration AddComponentConfiguration(
    System.string Name,
    System.string Comment,
    System.string AlternateName,
    System.int Options
)
```

C++/CLI

```
Configuration^ AddComponentConfiguration(
    & System.String^ Name,
    & System.String^ Comment,
    & System.String^ AlternateName,
    & System.int Options
)
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Parameters

*Name*  
Name of new configuration

*Comment*  
Comment displayed in **Configuration Properties**

*AlternateName*  
Alternate configuration name; used if swConfigOption\_UseAlternateName is set to true (see **Remarks**)

*Options*  
Combination of one or more BOOLEAN configuration options as defined in [swConfigurationOptions2](#) e (see **Remarks**)

#### Return Value

[IConfiguration](#)

### Visual Basic for Applications (VBA) Syntax

See [AssemblyDoc::AddComponentConfiguration](#).

### Remarks

The Options argument can be a combination of any of the following values:

- swConfigOption\_SuppressByDefault True if you want to suppress newly added features and mates in this configuration, false if not
- swConfigOption\_HideByDefault - True if you want newly added components to be hidden, false if not
- swConfigOption\_MinFeatureManager True if you want newly added components to only display their component name in the FeatureManager design tree, false if you want newly added components to display their name and each of their features in the FeatureManager design tree
- swConfigOption\_DontActivate - True if you do not want the new configuration activated, false if not

### See Also

[IAssemblyDoc Interface](#)  
[IAssemblyDoc Members](#)

### Availability

SOLIDWORKS 2005 FCS, Revision Number 13

SOLIDWORKS API Help

[Send comments on this topic.](#)

## AddComponents Method (IAssemblyDoc)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IAssemblyDoc Interface](#) : AddComponents Method (IAssemblyDoc)

Obsolete. Superseded by [IAssemblyDoc::AddComponents3](#).

### .NET Syntax

#### Visual Basic (Declaration)

```
Function AddComponents( _  
    ByVal Names As System.Object, _  
    ByVal Transforms As System.Object _  
) As System.Object
```

#### Visual Basic (Usage)



```
Dim instance As IAssemblyDoc  
Dim Names As System.Object  
Dim Transforms As System.Object  
Dim value As System.Object  
  
value = instance.AddComponents(Names, Transforms)
```

#### C#

```
System.object AddComponents(  
    System.object Names,  
    System.object Transforms  
)
```

#### C++/CLI

```
System.Object^ AddComponents(  
& System.Object^ Names,  
& System.Object^ Transforms  
)
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Parameters

*Names*

Array of strings component file names

*Transforms*

Array of doubles of the component transforms

#### Return Value

Array of objects of newly created components

### Visual Basic for Applications (VBA) Syntax

See [AssemblyDoc::AddComponents](#).

### Example

[Add Components \(VBA\)](#)

### Remarks

The array of file names represents all of the components that are added to the assembly. If there is more than one instance of a given component, make sure you add the file name of the component for each instance of the component.

The array of transforms consists of [count x 16] doubles. There should be one transform for each component being added.

**TIP:** See [ISldWorks::OpenDoc6](#) for tips on how to avoid using large amounts of memory when opening up multiple parts to add to an assembly

### See Also

[IAssemblyDoc Interface](#)

[IAssemblyDoc Members](#)

[IAssemblyDoc::IAddComponents2 Method](#)

### Availability

## AddComponents3 Method (IAssemblyDoc)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IAssemblyDoc Interface](#) : AddComponents3 Method (IAssemblyDoc)

Adds multiple components to the assembly.

### .NET Syntax

#### Visual Basic (Declaration)

```
Function AddComponents3( _
    ByVal Names As System.Object, _
    ByVal Transforms As System.Object, _
    ByVal CoordinateSystemNames As System.Object _
) As System.Object
```

#### Visual Basic (Usage)



```
Dim instance As IAssemblyDoc
Dim Names As System.Object
Dim Transforms As System.Object
Dim CoordinateSystemNames As System.Object
Dim value As System.Object

value = instance.AddComponents3(Names, Transforms, CoordinateSystemNames)
```

#### C#

```
System.object AddComponents3(
    System.object Names,
    System.object Transforms,
    System.object CoordinateSystemNames
)
```

#### C++/CLI

```
System.Object^ AddComponents3(
    & System.Object^ Names,
    & System.Object^ Transforms,
    & System.Object^ CoordinateSystemNames
)
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Parameters

*Names*  
Array of full path names of components  
*Transforms*  
Array of transformation matrix doubles  
*CoordinateSystemNames*  
Array of coordinate system names

#### Return Value

Array of [IComponent2](#) objects

### Visual Basic for Applications (VBA) Syntax

See [AssemblyDoc::AddComponents3](#).

### Example

[Add Components \(C#\)](#)  
[Add Components \(VB.NET\)](#)  
[Add Components \(VBA\)](#)

### Remarks

Transforms contains an array of [(Names count) x 16] doubles. This parameter stores one transformation matrix of 16 doubles for each component in Names. If a component's transformation matrix is null, then the component is placed in the assembly such that the component's

user-defined coordinate system coincides exactly with the default coordinate system of the assembly (no transformation). See [IMathTransform](#) for details about transformation matrices.

CoordinateSystemNames contains a user-defined coordinate system name for each component in Names. If a component's user-defined coordinate system is an empty string, then the component is placed in the assembly with respect to the default coordinate system of the component.

**TIP:** See [ISldWorks::OpenDoc6](#) for tips on how to avoid using large amounts of memory when opening up multiple parts to add to an assembly

## See Also

[IAssemblyDoc Interface](#)  
[IAssemblyDoc Members](#)  
[IAssemblyDoc::IAddComponents3 Method](#)

## Availability

SOLIDWORKS 2011 FCS, Revision Number 19.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

## AddConcentricMateWithTolerance Method (IAssemblyDoc)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IAssemblyDoc Interface](#) : AddConcentricMateWithTolerance Method (IAssemblyDoc)

Adds a misaligned concentric mate to this assembly.

## Visual Basic Syntax

Visual Basic (Declaration)

```
Function AddConcentricMateWithTolerance( _  
    ByVal AlignFromEnum As System.Integer, _  
    ByVal ConcentricPositionType As System.Integer, _  
    ByVal ConcentricToleranceCheck As System.Boolean, _  
    ByVal ConcentricToleranceValue As System.Double, _  
    ByRef ErrorStatus As System.Integer _  
    ) As Mate2
```

Visual Basic (Usage)



```
Dim instance As IAssemblyDoc  
Dim AlignFromEnum As System.Integer  
Dim ConcentricPositionType As System.Integer  
Dim ConcentricToleranceCheck As System.Boolean  
Dim ConcentricToleranceValue As System.Double  
Dim ErrorStatus As System.Integer  
Dim value As Mate2  
  
value = instance.AddConcentricMateWithTolerance(AlignFromEnum, ConcentricPositionType,  
ConcentricToleranceCheck, ConcentricToleranceValue, ErrorStatus)
```

C#

```
Mate2 AddConcentricMateWithTolerance(  
    System.int AlignFromEnum,  
    System.int ConcentricPositionType,  
    System.bool ConcentricToleranceCheck,  
    System.double ConcentricToleranceValue,  
    out System.int ErrorStatus  
)
```

C++/CLI

```
Mate2^ AddConcentricMateWithTolerance(  
& System.int AlignFromEnum,  
& System.int ConcentricPositionType,  
& System.bool ConcentricToleranceCheck,  
& System.double ConcentricToleranceValue,
```

```
& [Out] System.int ErrorStatus  
)
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Parameters

*AlignFromEnum*

Type of mate alignment as defined in [swMateAlign\\_e](#)

*ConcentricPositionType*

Misaligned concentric mate position as defined in [swConcentricAlignmentType\\_e](#)

*ConcentricToleranceCheck*

True to override the deviation, false to have SOLIDWORKS calculate the deviation

*ConcentricToleranceValue*

Maximum deviation; valid only when ConcentricToleranceCheck is true

*ErrorStatus*

Success or error as defined by [swAddMateError\\_e](#)

#### Return Value

Misaligned concentric [mate](#)

### Visual Basic for Applications (VBA) Syntax

See [AssemblyDoc::AddConcentricMateWithTolerance](#).

### Example

[Add and Edit Misaligned Symmetric Concentric Mate \(VBA\)](#)

[Add and Edit Misaligned Symmetric Concentric Mate \(VB.NET\)](#)

[Add and Edit Misaligned Symmetric Concentric Mate \(C#\)](#)

### Remarks

Select the two entities to mate before calling this method. A parent non-misaligned (i.e., standard) concentric mate must already exist between the selected entities. If a suitable parent concentric mate does not exist between the selected entities, then a standard concentric mate is created.

**NOTE:** The typical way to add a standard concentric mate is to use [IAssemblyDoc::AddMate5](#).

### See Also

[IAssemblyDoc Interface](#)

[IAssemblyDoc Members](#)

[IAssemblyDoc::EditConcentricMate Method \(\)](#)

[IAssemblyDoc::EditMate4 Method \(\)](#)

### Availability

SOLIDWORKS 2018 FCS, Revision Number 26.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

## AddDistanceMate Method (IAssemblyDoc)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IAssemblyDoc Interface](#) : AddDistanceMate Method (IAssemblyDoc)

Adds a distance mate to this assembly.

### .NET Syntax

Visual Basic (Declaration)

```
Function AddDistanceMate( _  
    ByVal AlignFromEnum As System.Integer, _  
    ByVal Flip As System.Boolean, _  
    ByVal Distance As System.Double, _  
    ByVal DistanceAbsUpperLimit As System.Double, _  
    ByVal DistanceAbsLowerLimit As System.Double, _  
    ByVal FirstArcCondition As System.Integer, _  
    ByVal SecondArcCondition As System.Integer, _  
    ByRef ErrorStatus As System.Integer _  
) As Mate2
```

Visual Basic (Usage)

```
Dim instance As IAssemblyDoc
Dim AlignFromEnum As System.Integer
Dim Flip As System.Boolean
Dim Distance As System.Double
Dim DistanceAbsUpperLimit As System.Double
Dim DistanceAbsLowerLimit As System.Double
Dim FirstArcCondition As System.Integer
Dim SecondArcCondition As System.Integer
Dim ErrorStatus As System.Integer
Dim value As Mate2

value = instance.AddDistanceMate(AlignFromEnum, Flip, Distance, DistanceAbsUpperLimit,
DistanceAbsLowerLimit, FirstArcCondition, SecondArcCondition, ErrorStatus)
```

C#

```
Mate2 AddDistanceMate(
    System.int AlignFromEnum,
    System.bool Flip,
    System.double Distance,
    System.double DistanceAbsUpperLimit,
    System.double DistanceAbsLowerLimit,
    System.int FirstArcCondition,
    System.int SecondArcCondition,
    out System.int ErrorStatus
)
```

C++/CLI

```
Mate2^ AddDistanceMate(
    & System.int AlignFromEnum,
    & System.bool Flip,
    & System.double Distance,
    & System.double DistanceAbsUpperLimit,
    & System.double DistanceAbsLowerLimit,
    & System.int FirstArcCondition,
    & System.int SecondArcCondition,
    & [Out] System.int ErrorStatus
)
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Parameters

*AlignFromEnum*  
Type of alignment as defined in [swMateAlign\\_e](#)

*Flip*  
True to flip mate entities, false to not (see **Remarks**)

*Distance*  
Distance value (see **Remarks**)

*DistanceAbsUpperLimit*  
Absolute maximum distance value (see **Remarks**)

*DistanceAbsLowerLimit*  
Absolute minimum distance value (see **Remarks**)

*FirstArcCondition*  
First arc condition as defined in [swDistanceMateArcConditions\\_e](#); valid only for cylindrical distance mates (see **Remarks**)

*SecondArcCondition*  
Second arc condition as defined in [swDistanceMateArcConditions\\_e](#); valid only for cylindrical distance mates (see **Remarks**)

*ErrorStatus*  
Success or error as defined by [swAddMateError\\_e](#)

#### Return Value

[IMate2](#)

### Visual Basic for Applications (VBA) Syntax

See [AssemblyDoc::AddDistanceMate](#).

### Example

[Add and Edit Distance Mate \(VBA\)](#)  
[Add and Edit Distance Mate \(VB.NET\)](#)  
[Add and Edit Distance Mate \(C#\)](#)

## Remarks

To specify a distance mate without limits, set the DistanceAbsUpperLimit and DistanceAbsLowerLimit parameters equal to the Distance parameter.

If the mate is applied to the closest position that meets the mate condition specified by Distance, then setting Flip to true moves the components to the other possible mate position.

For cylindrical distance mates only, the following FirstArcCondition-to-SecondArcCondition distance combinations are possible:

| FirstArcCondition as defined in <a href="#">swDistanceMateArcConditions_e</a> | to | SecondArcCondition as defined in <a href="#">swDistanceMateArcConditions_e</a> |
|-------------------------------------------------------------------------------|----|--------------------------------------------------------------------------------|
| swArcCondition_Center                                                         |    | swArcCondition_Center                                                          |
| swArcCondition_Center                                                         |    | swArcCondition_Minimum                                                         |
| swArcCondition_Center                                                         |    | swArcCondition_Maximum                                                         |
| swArcCondition_Minimum                                                        |    | swArcCondition_Center                                                          |
| swArcCondition_Minimum                                                        |    | swArcCondition_Minimum                                                         |
| swArcCondition_Minimum                                                        |    | swArcCondition_Maximum                                                         |
| swArcCondition_Maximum                                                        |    | swArcCondition_Center                                                          |
| swArcCondition_Maximum                                                        |    | swArcCondition_Minimum                                                         |
| swArcCondition_Maximum                                                        |    | swArcCondition_Maximum                                                         |

To add a distance mate:

1. Call [IModelDocExtension::SelectByRay](#) and [ISelectionMgr::GetSelectedObject6](#) to get each mate entity. (For cylindrical distance mates, the selections must be two cylindrical faces and one cylindrical face and one axis.)
2. Call [IModelDoc2::ClearSelection2](#).
3. Call [IEntity::Select4](#) to select each entity.
4. Call this method.
5. Call [IModelDoc2::EditRebuild3](#) after the mate is created.

If entities are not preselected, then ErrorStatus is swAddMateError\_e.swAddMateError\_IncorrectSelections, and nothing is returned.

## See Also

[IAssemblyDoc Interface](#)  
[IAssemblyDoc Members](#)  
[IAssemblyDoc::AddMate5 Method \(\)](#)  
[IAssemblyDoc::EditDistanceMate Method \(\)](#)  
[IMate2::DistanceFirstArcCondition Property \(\)](#)  
[IMate2::DistanceSecondArcCondition Property \(\)](#)

## Availability

SOLIDWORKS 2018 FCS, Revision Number 26.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

## AddMate Method (IAssemblyDoc)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IAssemblyDoc Interface](#) : AddMate Method (IAssemblyDoc)

Obsolete. Superseded by [IAssemblyDoc::AddMate3](#).

## .NET Syntax

Visual Basic (Declaration)

```
Sub AddMate( _  
    ByVal MateType As System.Integer, _  
    ByVal Align As System.Integer, _  
    ByVal Flip As System.Boolean, _  
    ByVal Dist As System.Double, _  
    ByVal Angle As System.Double _  
)
```

Visual Basic (Usage)

```
Dim instance As IAssemblyDoc  
Dim MateType As System.Integer  
Dim Align As System.Integer  
Dim Flip As System.Boolean  
Dim Dist As System.Double  
Dim Angle As System.Double
```



```
instance.AddMate(MateType, Align, Flip, Dist, Angle)
```

C#

```
void AddMate(  
    System.int MateType,  
    System.int Align,  
    System.bool Flip,  
    System.double Dist,  
    System.double Angle  
)
```

C++/CLI

```
void AddMate(  
& System.int MateType,  
& System.int Align,  
& System.bool Flip,  
& System.double Dist,  
& System.double Angle  
)
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Parameters

*MateType*

*Align*

*Flip*

*Dist*

*Angle*

### Visual Basic for Applications (VBA) Syntax

See [AssemblyDoc::AddMate](#).

### See Also

[IAssemblyDoc Interface](#)  
[IAssemblyDoc Members](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

## AddMate2 Method (IAssemblyDoc)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IAssemblyDoc Interface](#) : AddMate2 Method (IAssemblyDoc)

Obsolete. Superseded by [IAssemblyDoc::AddMate3](#).

### .NET Syntax

Visual Basic (Declaration)

```
Function AddMate2( _  
    ByVal MateTypeFromEnum As System.Integer, _  
    ByVal AlignFromEnum As System.Integer, _  
    ByVal Flip As System.Boolean, _  
    ByVal Distance As System.Double, _  
    ByVal DistanceAbsUpperLimit As System.Double, _  
    ByVal DistanceAbsLowerLimit As System.Double, _  
    ByVal GearRatioNumerator As System.Double, _  
    ByVal GearRatioDenominator As System.Double, _  
    ByVal Angle As System.Double, _  
    ByVal AngleAbsUpperLimit As System.Double, _  
    ByVal AngleAbsLowerLimit As System.Double, _  
    ByRef ErrorStatus As System.Integer _  
) As Mate2
```

```

Dim instance As IAssemblyDoc
Dim MateTypeFromEnum As System.Integer
Dim AlignFromEnum As System.Integer
Dim Flip As System.Boolean
Dim Distance As System.Double
Dim DistanceAbsUpperLimit As System.Double
Dim DistanceAbsLowerLimit As System.Double
Dim GearRatioNumerator As System.Double
Dim GearRatioDenominator As System.Double
Dim Angle As System.Double
Dim AngleAbsUpperLimit As System.Double
Dim AngleAbsLowerLimit As System.Double
Dim ErrorStatus As System.Integer
Dim value As Mate2

value = instance.AddMate2(MateTypeFromEnum, AlignFromEnum, Flip, Distance, DistanceAbsUpperLimit,
DistanceAbsLowerLimit, GearRatioNumerator, GearRatioDenominator, Angle, AngleAbsUpperLimit,
AngleAbsLowerLimit, ErrorStatus)

```

C#

```

Mate2 AddMate2(
    System.int MateTypeFromEnum,
    System.int AlignFromEnum,
    System.bool Flip,
    System.double Distance,
    System.double DistanceAbsUpperLimit,
    System.double DistanceAbsLowerLimit,
    System.double GearRatioNumerator,
    System.double GearRatioDenominator,
    System.double Angle,
    System.double AngleAbsUpperLimit,
    System.double AngleAbsLowerLimit,
    out System.int ErrorStatus
)

```

C++/CLI

```

Mate2^ AddMate2(
    & System.int MateTypeFromEnum,
    & System.int AlignFromEnum,
    & System.bool Flip,
    & System.double Distance,
    & System.double DistanceAbsUpperLimit,
    & System.double DistanceAbsLowerLimit,
    & System.double GearRatioNumerator,
    & System.double GearRatioDenominator,
    & System.double Angle,
    & System.double AngleAbsUpperLimit,
    & System.double AngleAbsLowerLimit,
    & [Out] System.int ErrorStatus
)

```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Parameters

*MateTypeFromEnum*

*AlignFromEnum*

*Flip*

*Distance*

*DistanceAbsUpperLimit*

*DistanceAbsLowerLimit*

*GearRatioNumerator*

*GearRatioDenominator*

*Angle*

*AngleAbsUpperLimit*

*AngleAbsLowerLimit*

*ErrorStatus*

## Visual Basic for Applications (VBA) Syntax

See [AssemblyDoc::AddMate2](#).

## See Also

[IAssemblyDoc Interface](#)  
[IAssemblyDoc Members](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

## AddMate3 Method (IAssemblyDoc)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IAssemblyDoc Interface](#) : AddMate3 Method (IAssemblyDoc)

Obsolete. Superseded by [IAssemblyDoc::AddMate4](#).

## .NET Syntax

Visual Basic (Declaration)

```
Function AddMate3( _  
    ByVal MateTypeFromEnum As System.Integer, _  
    ByVal AlignFromEnum As System.Integer, _  
    ByVal Flip As System.Boolean, _  
    ByVal Distance As System.Double, _  
    ByVal DistanceAbsUpperLimit As System.Double, _  
    ByVal DistanceAbsLowerLimit As System.Double, _  
    ByVal GearRatioNumerator As System.Double, _  
    ByVal GearRatioDenominator As System.Double, _  
    ByVal Angle As System.Double, _  
    ByVal AngleAbsUpperLimit As System.Double, _  
    ByVal AngleAbsLowerLimit As System.Double, _  
    ByVal ForPositioningOnly As System.Boolean, _  
    ByRef ErrorStatus As System.Integer _  
    ) As Mate2
```

Visual Basic (Usage)



```
Dim instance As IAssemblyDoc  
Dim MateTypeFromEnum As System.Integer  
Dim AlignFromEnum As System.Integer  
Dim Flip As System.Boolean  
Dim Distance As System.Double  
Dim DistanceAbsUpperLimit As System.Double  
Dim DistanceAbsLowerLimit As System.Double  
Dim GearRatioNumerator As System.Double  
Dim GearRatioDenominator As System.Double  
Dim Angle As System.Double  
Dim AngleAbsUpperLimit As System.Double  
Dim AngleAbsLowerLimit As System.Double  
Dim ForPositioningOnly As System.Boolean  
Dim ErrorStatus As System.Integer  
Dim value As Mate2
```

```
value = instance.AddMate3(MateTypeFromEnum, AlignFromEnum, Flip, Distance, DistanceAbsUpperLimit,  
DistanceAbsLowerLimit, GearRatioNumerator, GearRatioDenominator, Angle, AngleAbsUpperLimit,  
AngleAbsLowerLimit, ForPositioningOnly, ErrorStatus)
```

C#

```
Mate2 AddMate3(  
    System.int MateTypeFromEnum,
```

```

System.int AlignFromEnum,
System.bool Flip,
System.double Distance,
System.double DistanceAbsUpperLimit,
System.double DistanceAbsLowerLimit,
System.double GearRatioNumerator,
System.double GearRatioDenominator,
System.double Angle,
System.double AngleAbsUpperLimit,
System.double AngleAbsLowerLimit,
System.bool ForPositioningOnly,
out System.int ErrorStatus
)

```

C++/CLI

```

Mate2^ AddMate3(
& System.int MateTypeFromEnum,
& System.int AlignFromEnum,
& System.bool Flip,
& System.double Distance,
& System.double DistanceAbsUpperLimit,
& System.double DistanceAbsLowerLimit,
& System.double GearRatioNumerator,
& System.double GearRatioDenominator,
& System.double Angle,
& System.double AngleAbsUpperLimit,
& System.double AngleAbsLowerLimit,
& System.bool ForPositioningOnly,
[Out] System.int ErrorStatus
)

```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Parameters

*MateTypeFromEnum*

Type of mate as defined in [swMateType\\_e](#) (see **Remarks**)

*AlignFromEnum*

Type of alignment as defined in [swMateAlign\\_e](#)

*Flip*

True to flip the component, false otherwise

*Distance*

Distance value to use with distance or limit mates

*DistanceAbsUpperLimit*

Absolute maximum distance value (see **Remarks**)

*DistanceAbsLowerLimit*

Absolute minimum distance value (see **Remarks**)

*GearRatioNumerator*

Gear ratio numerator value for gear mates

*GearRatioDenominator*

Gear ratio denominator value for gear mates

*Angle*

Angle value to use with angle mates

*AngleAbsUpperLimit*

Absolute maximum angle value

*AngleAbsLowerLimit*

Absolute maximum angle value

*ForPositioningOnly*

True to only position the components according to the mating relationship and not create and return a mate, false to not

*ErrorStatus*

Success or error as defined by [swAddMateError\\_e](#)

#### Return Value

[IMate2](#)

## Visual Basic for Applications (VBA) Syntax

See [AssemblyDoc::AddMate3](#).

## Remarks

To specify a distance mate without limits, set the DistanceAbsUpperLimit and DistanceAbsLowerLimit arguments equal to the distance argument's value.

If MateTypeFromEnum is swMateDISTANCE or swMateANGLE when the mate is applied to the closest position that meets the mate condition specified by distance or angle, then setting Flip to true moves the assembly to the other possible mate position.

To add a mate:

1. Call [IModelDoc2::ClearSelection2](#) before selecting entities to mate.
2. Call [IModelDocExtension::SelectByID2](#) to select entities to mate.  
If MateTypeFromEnum is swMateCAMFOLLOWER, then use a selection mark of 8 for the cam-follower face.  
If MateTypeFromEnum is swMateWIDTH and uses tab selections, then use a selection mark of 16. If MateTypeFromEnum is anything else, then use a selection mark of 1.
3. Call this method.
4. Call [IModelDoc2::ClearSelection2](#) after the mate is created.

If nothing is preselected, then ErrorStatus is swAddMateError\_IncorrectSelections, and nothing is returned.

## See Also

[IAssemblyDoc Interface](#)  
[IAssemblyDoc Members](#)  
[IAssemblyDoc::EditMate2 Method](#)  
[IAssemblyDoc::CopyWithMates Method](#)

## Availability

SOLIDWORKS 2005 SP2, Revision Number 13

SOLIDWORKS API Help

[Send comments on this topic.](#)

## AddMate4 Method (IAssemblyDoc)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IAssemblyDoc Interface](#) : AddMate4 Method (IAssemblyDoc)

Obsolete. Superseded by [IAssemblyDoc::AddMate5](#).

## Visual Basic Syntax

Visual Basic (Declaration)

```
Function AddMate4( _  
    ByVal MateTypeFromEnum As System.Integer, _  
    ByVal AlignFromEnum As System.Integer, _  
    ByVal Flip As System.Boolean, _  
    ByVal Distance As System.Double, _  
    ByVal DistanceAbsUpperLimit As System.Double, _  
    ByVal DistanceAbsLowerLimit As System.Double, _  
    ByVal GearRatioNumerator As System.Double, _  
    ByVal GearRatioDenominator As System.Double, _  
    ByVal Angle As System.Double, _  
    ByVal AngleAbsUpperLimit As System.Double, _  
    ByVal AngleAbsLowerLimit As System.Double, _  
    ByVal ForPositioningOnly As System.Boolean, _  
    ByVal LockRotation As System.Boolean, _  
    ByRef ErrorStatus As System.Integer _  
) As Mate2
```

Visual Basic (Usage)



```
Dim instance As IAssemblyDoc  
Dim MateTypeFromEnum As System.Integer  
Dim AlignFromEnum As System.Integer  
Dim Flip As System.Boolean  
Dim Distance As System.Double  
Dim DistanceAbsUpperLimit As System.Double  
Dim DistanceAbsLowerLimit As System.Double  
Dim GearRatioNumerator As System.Double  
Dim GearRatioDenominator As System.Double  
Dim Angle As System.Double
```

```

Dim AngleAbsUpperLimit As System.Double
Dim AngleAbsLowerLimit As System.Double
Dim ForPositioningOnly As System.Boolean
Dim LockRotation As System.Boolean
Dim ErrorStatus As System.Integer
Dim value As Mate2

```

```

value = instance.AddMate4(MateTypeFromEnum, AlignFromEnum, Flip, Distance, DistanceAbsUpperLimit,
DistanceAbsLowerLimit, GearRatioNumerator, GearRatioDenominator, Angle, AngleAbsUpperLimit,
AngleAbsLowerLimit, ForPositioningOnly, LockRotation, ErrorStatus)

```

C#

```

Mate2 AddMate4(
    System.int MateTypeFromEnum,
    System.int AlignFromEnum,
    System.bool Flip,
    System.double Distance,
    System.double DistanceAbsUpperLimit,
    System.double DistanceAbsLowerLimit,
    System.double GearRatioNumerator,
    System.double GearRatioDenominator,
    System.double Angle,
    System.double AngleAbsUpperLimit,
    System.double AngleAbsLowerLimit,
    System.bool ForPositioningOnly,
    System.bool LockRotation,
    out System.int ErrorStatus
)

```

C++/CLI

```

Mate2^ AddMate4(
    & System.int MateTypeFromEnum,
    & System.int AlignFromEnum,
    & System.bool Flip,
    & System.double Distance,
    & System.double DistanceAbsUpperLimit,
    & System.double DistanceAbsLowerLimit,
    & System.double GearRatioNumerator,
    & System.double GearRatioDenominator,
    & System.double Angle,
    & System.double AngleAbsUpperLimit,
    & System.double AngleAbsLowerLimit,
    & System.bool ForPositioningOnly,
    & System.bool LockRotation,
    & [Out] System.int ErrorStatus
)

```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

## Parameters

*MateTypeFromEnum*

Type of mate as defined in [swMateType\\_e](#) (see **Remarks**)

*AlignFromEnum*

Type of alignment as defined in [swMateAlign\\_e](#)

*Flip*

True to flip the component, false otherwise

*Distance*

Distance value to use with distance or limit mates

*DistanceAbsUpperLimit*

Absolute maximum distance value (see **Remarks**)

*DistanceAbsLowerLimit*

Absolute minimum distance value (see **Remarks**)

*GearRatioNumerator*

Gear ratio numerator value for gear mates

*GearRatioDenominator*

Gear ratio denominator value for gear mates

*Angle*

Angle value to use with angle mates

*AngleAbsUpperLimit*

Absolute maximum angle value

*AngleAbsLowerLimit*

Absolute minimum angle value

*ForPositioningOnly*

True to only position the components according to the mating relationship and not return a mate, false to return a mate

*LockRotation*

True to lock component rotation, false to not

*ErrorStatus*

Success or error as defined by [swAddMateError\\_e](#)

## Visual Basic for Applications (VBA) Syntax

See [AssemblyDoc::AddMate4](#).

## Remarks

The difference between this method and the now obsolete `IAssemblyDoc::AddMate3` is that this method's `LockRotation` parameter provides the option to lock the rotation of components in the mate.

To specify a distance mate without limits, set the `DistanceAbsUpperLimit` and `DistanceAbsLowerLimit` parameters equal to the `Distance` parameter.

If `MateTypeFromEnum` is `swMateType_e.swMateDISTANCE` or `swMateType_e.swMateANGLE`, and the mate is applied to the closest position that meets the mate condition specified by `Distance` or `Angle`, then setting `Flip` to true moves the assembly to the other possible mate position.

To add a mate:

1. Call [IModelDoc2::ClearSelection2](#) before selecting entities to mate.
2. Call [IModelDocExtension::SelectByID2](#) to select each entity to mate.

| If <code>MateTypeFromEnum</code> is...      | Use selection mark... |
|---------------------------------------------|-----------------------|
| <code>swMateType_e.swMateCAMFOLLOWER</code> | 8                     |
| <code>swMateType_e.swMateWIDTH</code>       | 16                    |
| Other <code>swMateType_e_option</code>      | 1                     |

3. Call this method.
4. Call [IModelDoc2::ClearSelection2](#) after the mate is created.

If nothing is preselected, then `ErrorStatus` is `swAddMateError_IncorrectSelections`, and nothing is returned.

## See Also

[IAssemblyDoc Interface](#)  
[IAssemblyDoc Members](#)

## Availability

SOLIDWORKS 2014 FCS, Revision Number 22.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

## AddMate5 Method (IAssemblyDoc)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IAssemblyDoc Interface](#) : AddMate5 Method (IAssemblyDoc)

Obsolete. Superseded by [IAssemblyDoc::CreateMate](#).

## .NET Syntax

Visual Basic (Declaration)

```
Function AddMate5( _  
    ByVal MateTypeFromEnum As System.Integer, _  
    ByVal AlignFromEnum As System.Integer, _  
    ByVal Flip As System.Boolean, _  
    ByVal Distance As System.Double, _  
    ByVal DistanceAbsUpperLimit As System.Double, _  
    ByVal DistanceAbsLowerLimit As System.Double, _  
    ByVal GearRatioNumerator As System.Double, _  
    ByVal GearRatioDenominator As System.Double, _  
    ByVal Angle As System.Double, _  
    ByVal AngleAbsUpperLimit As System.Double, _  
    ByVal AngleAbsLowerLimit As System.Double, _  
    ByVal ForPositioningOnly As System.Boolean, _  
    ByVal LockRotation As System.Boolean, _  
    ByVal WidthMateOption As System.Integer, _  
    ByRef ErrorStatus As System.Integer _  
) As Mate2
```

Visual Basic (Usage)



```

Dim instance As IAssemblyDoc
Dim MateTypeFromEnum As System.Integer
Dim AlignFromEnum As System.Integer
Dim Flip As System.Boolean
Dim Distance As System.Double
Dim DistanceAbsUpperLimit As System.Double
Dim DistanceAbsLowerLimit As System.Double
Dim GearRatioNumerator As System.Double
Dim GearRatioDenominator As System.Double
Dim Angle As System.Double
Dim AngleAbsUpperLimit As System.Double
Dim AngleAbsLowerLimit As System.Double
Dim ForPositioningOnly As System.Boolean
Dim LockRotation As System.Boolean
Dim WidthMateOption As System.Integer
Dim ErrorStatus As System.Integer
Dim value As Mate2

```

```

value = instance.AddMate5(MateTypeFromEnum, AlignFromEnum, Flip, Distance, DistanceAbsUpperLimit,
DistanceAbsLowerLimit, GearRatioNumerator, GearRatioDenominator, Angle, AngleAbsUpperLimit,
AngleAbsLowerLimit, ForPositioningOnly, LockRotation, WidthMateOption, ErrorStatus)

```

C#

```

Mate2 AddMate5(
    System.int MateTypeFromEnum,
    System.int AlignFromEnum,
    System.bool Flip,
    System.double Distance,
    System.double DistanceAbsUpperLimit,
    System.double DistanceAbsLowerLimit,
    System.double GearRatioNumerator,
    System.double GearRatioDenominator,
    System.double Angle,
    System.double AngleAbsUpperLimit,
    System.double AngleAbsLowerLimit,
    System.bool ForPositioningOnly,
    System.bool LockRotation,
    System.int WidthMateOption,
    out System.int ErrorStatus
)

```

C++/CLI

```

Mate2^ AddMate5(
    & System.int MateTypeFromEnum,
    & System.int AlignFromEnum,
    & System.bool Flip,
    & System.double Distance,
    & System.double DistanceAbsUpperLimit,
    & System.double DistanceAbsLowerLimit,
    & System.double GearRatioNumerator,
    & System.double GearRatioDenominator,
    & System.double Angle,
    & System.double AngleAbsUpperLimit,
    & System.double AngleAbsLowerLimit,
    & System.bool ForPositioningOnly,
    & System.bool LockRotation,
    & System.int WidthMateOption,
    & [Out] System.int ErrorStatus
)

```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Parameters

*MateTypeFromEnum*

Type of mate as defined in [swMateType\\_e](#) (see **Remarks**)

*AlignFromEnum*

Type of alignment as defined in [swMateAlign\\_e](#)

*Flip*

True to flip the mate entities, false to not; valid only if MateTypeFromEnum is swMatetype\_e.swMateDISTANCE

*Distance*  
Distance value; valid only if MateTypeFromEnum is swMateType\_e.swMateDISTANCE  
*DistanceAbsUpperLimit*  
Absolute maximum distance value; valid only if MateTypeFromEnum is swMateType\_e.swMateDISTANCE (see **Remarks**)  
*DistanceAbsLowerLimit*  
Absolute minimum distance value; valid only if MateTypeFromEnum is swMateType\_e.swMateDISTANCE (see **Remarks**)  
*GearRatioNumerator*  
Gear ratio numerator value; valid only if MateTypeFromEnum is swMateType\_e.swMateGEAR  
*GearRatioDenominator*  
Gear ratio denominator value; valid only if MateTypeFromEnum is swMateType\_e.swMateGEAR  
*Angle*  
Angle value; valid only if MateTypeFromEnum is swMateType\_e.swMateANGLE  
*AngleAbsUpperLimit*  
Absolute maximum angle value; valid only if MateTypeFromEnum is swMateType\_e.swMateANGLE  
*AngleAbsLowerLimit*  
Absolute minimum angle value; valid only if MateTypeFromEnum is swMateType\_e.swMateANGLE  
*ForPositioningOnly*  
True to only position the components according to the mating relationship and not return a mate, false to return a mate  
*LockRotation*  
True to lock component rotation, false to not  
*WidthMateOption*  
Width mate options as defined in [swMateWidthOptions\\_e](#); valid only if MateTypeFromEnum is swMateType\_e.swMateWIDTH  
*ErrorStatus*  
Success or error as defined by [swAddMateError\\_e](#)

**Return Value**

[Mate](#)

**Visual Basic for Applications (VBA) Syntax**

See [AssemblyDoc::AddMate5](#).

**Example**

[Add and Mate Component \(VBA\)](#)  
[Add and Mate Component \(VB.NET\)](#)  
[Add and Mate Component \(C#\)](#)  
[Copy Component With Profile Center Mate \(C#\)](#)  
[Copy Component With Profile Center Mate \(VB.NET\)](#)  
[Copy Component With Profile Center Mate \(VBA\)](#)

**Remarks**

To create:

- Standard, advanced, and mechanical mates, use [IAssemblyDoc::CreateMate](#).
- Misaligned concentric mates, use [IAssemblyDoc::AddConcentricMateWithTolerance](#).
- Cylindrical distance mates, use [IAssemblyDoc::AddDistanceMate](#).

The difference between this method and the now obsolete IAssemblyDoc::AddMate4 is the WidthMateOption parameter.

To specify a distance mate without limits, set the DistanceAbsUpperLimit and DistanceAbsLowerLimit parameters equal to the Distance parameter.

If MateTypeFromEnum is swMateType\_e.swMateDISTANCE or swMateType\_e.swMateANGLE, and the mate is applied to the closest position that meets the mate condition specified by Distance or Angle, then setting Flip to true moves the components to the other possible mate position

To add a standard or mechanical mate:

1. Call [IModelDoc2::ClearSelection2](#) before selecting entities to mate.
2. Call [IModelDocExtension::SelectByID2](#) to select each entity to mate.

| If MateTypeFromEnum is...      | Use selection mark... |
|--------------------------------|-----------------------|
| swMateType_e.swMateCAMFOLLOWER | 8                     |
| swMateType_e.swMateWIDTH       | 16                    |
| Other swMateType_e_option      | 1                     |

3. Call this method.
4. Call IModelDoc2::ClearSelection2 after the mate is created.

If entities are not preselected, then ErrorStatus is swAddMateError\_e.swAddMateError\_IncorrectSelections, and nothing is returned.

**See Also**

[IAssemblyDoc Interface](#)  
[IAssemblyDoc Members](#)  
[IAssemblyDoc::EditMate3 Method \(\)](#)  
[IAssemblyDoc::EditConcentricMate Method \(\)](#)  
[IAssemblyDoc::EditDistanceMate Method \(\)](#)

**Availability**

SOLIDWORKS 2015 FCS, Revision Number 23.0

# AddPipePenetration Method (IAssemblyDoc)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IAssemblyDoc Interface](#) : AddPipePenetration Method (IAssemblyDoc)

Penetrates the adjacent fitting or pipe with the pipe that ends at the selected sketch point.

## .NET Syntax

Visual Basic (Declaration)

```
Function AddPipePenetration() As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IAssemblyDoc  
Dim value As System.Integer  
  
value = instance.AddPipePenetration()
```

C#

```
System.int AddPipePenetration()
```

C++/CLI

```
System.int AddPipePenetration();
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

### Return Value

Indicates the success or failure of the operation as defined in [swPipingPenetrationStatus\\_e](#)

## Visual Basic for Applications (VBA) Syntax

See [AssemblyDoc::AddPipePenetration](#).

## Remarks

If the routing DLL is not available, then COM returns ITF\_E\_ROUTINGNOTLOADED.

## See Also

[IAssemblyDoc Interface](#)  
[IAssemblyDoc Members](#)  
[IAssemblyDoc::AddPipingFitting Method](#)

## Availability

SOLIDWORKS 99, datecode 1999207

SOLIDWORKS API Help

[Send comments](#) on this topic.

# AddPipingFitting Method (IAssemblyDoc)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IAssemblyDoc Interface](#) : AddPipingFitting Method (IAssemblyDoc)

Adds a pipe fitting to the current piping assembly.

## .NET Syntax

Visual Basic (Declaration)

```
Function AddPipingFitting( _  
    ByVal PathName As System.String, _  
    ByVal ConfigName As System.String, _
```

```
ByVal AlignmentIndex As System.Short _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IAssemblyDoc  
Dim PathName As System.String  
Dim ConfigName As System.String  
Dim AlignmentIndex As System.Short  
Dim value As System.Boolean
```

```
value = instance.AddPipingFitting(PathName, ConfigName, AlignmentIndex)
```

C#

```
System.bool AddPipingFitting(  
    System.string PathName,  
    System.string ConfigName,  
    System.short AlignmentIndex  
)
```

C++/CLI

```
System.bool AddPipingFitting(  
& System.String^ PathName,  
& System.String^ ConfigName,  
& System.short AlignmentIndex  
)
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Parameters

*PathName*

Full name and directory location of part file used for this fitting

*ConfigName*

Configuration within the fitting part file which should be used

*AlignmentIndex*

Each fitting has a varying number of alignment positions; this value allows you to choose the alignment position

#### Return Value

True if successful, false otherwise

### Visual Basic for Applications (VBA) Syntax

See [AssemblyDoc::AddPipingFitting](#).

### Remarks

This method adds a piping fitting to the selected sketch point. The sketch must be the active sketch of the piping assembly.

The alignmentIndex argument controls the alignment of the fitting. For example, you can align a t-piece in two ways: passing 0 aligns the t-piece one way, and passing 1 aligns it the other way.

If the routing DLL is not available, then COM returns ITF\_E\_ROUTINGNOTLOADED.

### See Also

[IAssemblyDoc Interface](#)

[IAssemblyDoc Members](#)

[IAssemblyDoc::AddPipePenetration Method](#)

### Availability

SOLIDWORKS 99, datecode 1999207



[SolidWorks.Interop.sldworks Namespace](#) > [IAssemblyDoc Interface](#) : AddPLMComponent Method (IAssemblyDoc)

Adds the specified component from a 3DEXPERIENCE collaborative space to the specified location in this assembly in [SOLIDWORKS Design connected to the 3DEXPERIENCE platform](#).

## .NET Syntax

### Visual Basic (Declaration)

```
Function AddPLMComponent( _  
    ByVal PLMID As System.String, _  
    ByVal ConfigName As System.String, _  
    ByVal X As System.Double, _  
    ByVal Y As System.Double, _  
    ByVal Z As System.Double _  
) As System.Object
```

### Visual Basic (Usage)



```
Dim instance As IAssemblyDoc  
Dim PLMID As System.String  
Dim ConfigName As System.String  
Dim X As System.Double  
Dim Y As System.Double  
Dim Z As System.Double  
Dim value As System.Object  
  
value = instance.AddPLMComponent(PLMID, ConfigName, X, Y, Z)
```

### C#

```
System.object AddPLMComponent(  
    System.string PLMID,  
    System.string ConfigName,  
    System.double X,  
    System.double Y,  
    System.double Z  
)
```

### C++/CLI

```
System.Object^ AddPLMComponent(  
& System.String^ PLMID,  
& System.String^ ConfigName,  
& System.double X,  
& System.double Y,  
& System.double Z  
)
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

### Parameters

*PLMID*

Unique ID of a component

*ConfigName*

Name of the configuration (Physical Product) from which to load the component

*X*

X-coordinate of the component center

*Y*

Y-coordinate of the component center

*Z*

Z-coordinate of the component center

### Return Value

[IComponent2](#)

## Visual Basic for Applications (VBA) Syntax

See [AssemblyDoc::AddPLMComponent](#).

## See Also

[IAssemblyDoc Interface](#)  
[IAssemblyDoc Members](#)  
[IAssemblyDoc::ReplacePLMComponents Method \(\)](#)

## Availability

SOLIDWORKS 2022 FCS, Revision Number 30

SOLIDWORKS API Help

[Send comments on this topic.](#)

## AddSmartComponent Method (IAssemblyDoc)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IAssemblyDoc Interface](#) : AddSmartComponent Method (IAssemblyDoc)

Adds the specified component at the specified coordinates as a Smart Component to this assembly.

## .NET Syntax

### Visual Basic (Declaration)

```
Function AddSmartComponent( _  
    ByVal CompName As System.String, _  
    ByVal X As System.Double, _  
    ByVal Y As System.Double, _  
    ByVal Z As System.Double _  
) As Component2
```

### Visual Basic (Usage)



```
Dim instance As IAssemblyDoc  
Dim CompName As System.String  
Dim X As System.Double  
Dim Y As System.Double  
Dim Z As System.Double  
Dim value As Component2  
  
value = instance.AddSmartComponent(CompName, X, Y, Z)
```

### C#

```
Component2 AddSmartComponent(  
    System.string CompName,  
    System.double X,  
    System.double Y,  
    System.double Z  
)
```

### C++/CLI

```
Component2^ AddSmartComponent(  
& System.String^ CompName,  
& System.double X,  
& System.double Y,  
& System.double Z  
)
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

### Parameters

*CompName*  
Path and filename of the part document to add as a Smart Component  
*X*  
X coordinate for the Smart Component  
*Y*  
Y coordinate for the Smart Component  
*Z*  
Z coordinate for the Smart Component

### Return Value

[Smart Component](#)

## Visual Basic for Applications (VBA) Syntax

See [AssemblyDoc::AddSmartComponent](#).

## See Also

[IAssemblyDoc Interface](#)  
[IAssemblyDoc Members](#)  
[IAssemblyDoc::CreateSmartComponent Method](#)  
[IComponent2::GetSmartComponentData Method](#)  
[IComponent2::IsSmartComponent Method](#)  
[IComponent2::SetSmartComponentData Method](#)  
[ISmartComponentFeatureData Interface](#)

## Availability

SOLIDWORKS 2010 FCS, Revision Number 18.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

## AddToFeatureScope Method (IAssemblyDoc)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IAssemblyDoc Interface](#) : AddToFeatureScope Method (IAssemblyDoc)

Adds a component to the scope of the currently selected assembly feature.

## .NET Syntax

### Visual Basic (Declaration)

```
Function AddToFeatureScope( _  
    ByVal CompName As System.String _  
) As System.Boolean
```

### Visual Basic (Usage)



```
Dim instance As IAssemblyDoc  
Dim CompName As System.String  
Dim value As System.Boolean  
  
value = instance.AddToFeatureScope(CompName)
```

### C#

```
System.bool AddToFeatureScope(  
    System.string CompName  
)
```

### C++/CLI

```
System.bool AddToFeatureScope(  
& System.String^ CompName  
)
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

### Parameters

*CompName*

Name of component to add

### Return Value

True if added successfully, false otherwise

## Visual Basic for Applications (VBA) Syntax

See [AssemblyDoc::AddToFeatureScope](#).

## Remarks

If no assembly feature is selected, then this method adds a component to a list for use with the next assembly feature created.

Assembly features allow you to create a feature that affects multiple components in an assembly. For example, if you need a hole through two blocks so that you could bolt them together, you can create a hole as an assembly feature and make it go through both blocks.

The scope of the assembly feature determines which components are effected by this feature. In other words, it describes which components contain the feature.

There are two ways to use the `IAssemblyDoc::AddToFeatureScope` method:

- When an assembly feature is selected, `IAssemblyDoc::AddToFeatureScope` adds a component to the assembly feature's scope. Any component in the scope of the assembly feature can be affected by the assembly feature.
- You can also use `IAssemblyDoc::AddToFeatureScope` to compile a group of components. The next assembly feature created (for example, [IFeatureManager::SimpleHole](#)) affects each of the components in the compiled group.

After completing these steps, use [IAssemblyDoc::UpdateFeatureScope](#) to display the changes.

## See Also

[IAssemblyDoc Interface](#)  
[IAssemblyDoc Members](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

## AssemblyPartToggle Method (IAssemblyDoc)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IAssemblyDoc Interface](#) : AssemblyPartToggle Method (IAssemblyDoc)

Obsolete. Superseded by [IAssemblyDoc::EditAssembly](#) and [IAssemblyDoc::EditPart2](#).

## .NET Syntax

Visual Basic (Declaration)

```
Sub AssemblyPartToggle()
```

Visual Basic (Usage)



```
Dim instance As IAssemblyDoc  
instance.AssemblyPartToggle()
```

C#

```
void AssemblyPartToggle()
```

C++/CLI

```
void AssemblyPartToggle();
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

## Visual Basic for Applications (VBA) Syntax

See [AssemblyDoc::AssemblyPartToggle](#).

## See Also

[IAssemblyDoc Interface](#)  
[IAssemblyDoc Members](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

## AutoAngleAxis Method (IAssemblyDoc)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IAssemblyDoc Interface](#) : AutoAngleAxis Method (IAssemblyDoc)

Automatically detect the axis for an angle mate.

## .NET Syntax

Visual Basic (Declaration)

```
Sub AutoAngleAxis( _  
    ByVal Mate As System.Object _  
)
```

Visual Basic (Usage)



```
Dim instance As IAssemblyDoc  
Dim Mate As System.Object  
  
instance.AutoAngleAxis(Mate)
```

C#

```
void AutoAngleAxis(  
    System.object Mate  
)
```

C++/CLI

```
void AutoAngleAxis(  
& System.Object^ Mate  
)
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Parameters

*Mate*  
[IMate2](#)

### Visual Basic for Applications (VBA) Syntax

See [AssemblyDoc::AutoAngleAxis](#).

### See Also

[IAssemblyDoc Interface](#)  
[IAssemblyDoc Members](#)

### Availability

SOLIDWORKS 2015 FCS, Revision Number 23.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

## AutoExplode Method (IAssemblyDoc)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IAssemblyDoc Interface](#) : AutoExplode Method (IAssemblyDoc)

Automatically generates an exploded view of the current assembly configuration.

### .NET Syntax

Visual Basic (Declaration)

```
Function AutoExplode() As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IAssemblyDoc  
Dim value As System.Boolean  
  
value = instance.AutoExplode()
```

C#

```
System.bool AutoExplode()
```

C++/CLI

```
System.bool AutoExplode();
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Return Value

True if the exploded view is automatically generated, false if not

### Visual Basic for Applications (VBA) Syntax

See [AssemblyDoc::AutoExplode](#).

### Example

See the [IExplodeStep](#) examples.

### Example

[Access Assembly \(C++\)](#)  
[Insert Explode Line Sketch and Route Line \(VB.NET\)](#)  
[Insert Explode Line Sketch and Route Line \(VBA\)](#)  
[Insert Explode Line Sketch and Route Line \(C#\)](#)

### Remarks

This method creates only one exploded view per assembly configuration. To create multiple exploded views, use [IAssemblyDoc::CreateExplodedView](#).

It is difficult to get the exploded view perfect without some user interaction. However, this method makes an attempt based on the current mate conditions.

### See Also

[IAssemblyDoc Interface](#)  
[IAssemblyDoc Members](#)  
[IAssemblyDoc::GetExplodedViewConfigurationName Method \(\)](#)  
[IAssemblyDoc::GetExplodedViewCount2 Method \(\)](#)  
[IAssemblyDoc::GetExplodedViewNames2 Method \(\)](#)  
[IAssemblyDoc::ShowExploded2 Method \(\)](#)  
[IAssemblyDoc::ViewCollapseAssembly Method \(\)](#)  
[IAssemblyDoc::ViewExplodeAssembly Method \(\)](#)  
[IModelDocExtension::IsExploded Method \(\)](#)  
[ISketchManager::InsertExplodeLineSketch Method \(\)](#)  
[IView::IsExploded Method \(\)](#)  
[IView::ShowExploded Method \(\)](#)  
[IConfiguration::AddRadialExplodeStep Method \(\)](#)  
[IConfiguration::AddExplodeStep2 Method \(\)](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

## AutoMateRepair Method (IAssemblyDoc)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IAssemblyDoc Interface](#) : AutoMateRepair Method (IAssemblyDoc)

Automatically repairs broken mates in this assembly.

### .NET Syntax

Visual Basic (Declaration)

```
Function AutoMateRepair( _  
    ByRef ProcessedMates As System.Object, _  
    ByRef FailedMates As System.Object _  
) As System.Integer
```

Visual Basic (Usage)

```
Dim instance As IAssemblyDoc  
Dim ProcessedMates As System.Object  
Dim FailedMates As System.Object  
Dim value As System.Integer
```



```
value = instance.AutoMateRepair(ProcessedMates, FailedMates)
```

C#

```
System.int AutoMateRepair(  
    out System.object ProcessedMates,  
    out System.object FailedMates  
)
```

C++/CLI

```
System.int AutoMateRepair(  
& [Out] System.Object^ ProcessedMates,  
& [Out] System.Object^ FailedMates  
)
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Parameters

*ProcessedMates*  
Array of successfully repaired [IMate2s](#)  
*FailedMates*  
Array of failed IMate2s

#### Return Value

Mate repair errors as defined by [swAutoMateRepairErrors\\_e](#)

### Visual Basic for Applications (VBA) Syntax

See [AssemblyDoc::AutoMateRepair](#).

### Example

'VBA

'-----

' Preconditions:

- ' 1. Open the Immediate window.
- ' 2. Ensure that **c:\temp** and the specified templates exist.

'

' Postconditions:

- ' 1. When the macro stops, inspect the FeatureManager design tree.
  - ' Observe that two Cylinder\_225 components, two planes, and two mates are added to the assembly.
- ' 2. In the FeatureManager design tree delete PLANE2.
  - ' Observe that Cylinder\_225<1> and Coincident2 are red, indicating a mate error has occurred.
- ' 3. In the user interface add a reference plane that is 1 mm from the Front Plane.
- ' 4. In the FeatureManager design tree select the Mates folder.
- ' 5. Press F5 to autorepair the broken mate.
  - ' The Immediate window shows one repaired mate.

'-----

```
Dim swApp As SldWorks.SldWorks  
Dim Part As SldWorks.ModelDoc2  
Dim boolstatus As Boolean  
Dim longstatus As Long, longwarnings As Long  
Option Explicit  
Sub main()
```

```

Set swApp = Application.SldWorks
Dim swSheetWidth As Double
swSheetWidth = 0
Dim swSheetHeight As Double
swSheetHeight = 0
Set Part = swApp.NewDocument("c:\temp\Part.prtdot", 0, swSheetWidth,
swSheetHeight)
Dim swPart As SldWorks.PartDoc
Set swPart = Part
swApp.ActivateDoc2 "Part1", False, longstatus
Set Part = swApp.ActiveDoc
Dim myModelView As SldWorks.ModelView
Set myModelView = Part.ActiveView
myModelView.FrameState = swWindowState_e.swWindowMaximized
Part.SketchManager.InsertSketch True
boolstatus = Part.Extension.SelectByID2("Front Plane", "PLANE",
-0.068513658157061, 4.03895355082156E-02, 8.71366929740688E-03, False, 0, Nothing, 0)
Part.ClearSelection2 True
Dim skSegment As Object
Set skSegment = Part.SketchManager.CreateCircle(0#, 0#, 0#, 0.045995, 0.021666,
0#)
Part.ShowNamedView2 "**Trimetric", 8
Part.ViewZoomtofit2
Dim myFeature As SldWorks.Feature
Set myFeature = Part.FeatureManager.FeatureExtrusion2(True, False, False, 0, 0,
0.225, 0.01, False, False, False, False, 1.74532925199433E-02, 1.74532925199433E-02,
False, False, False, False, True, True, True, 0, 0, False)
Part.SelectionManager.EnableContourSelection = False

' Save As
longstatus = Part.SaveAs3("c:\temp\Cylinder_225.SLDPRT", 0, 0)
swSheetWidth = 0
swSheetHeight = 0
Set Part = swApp.NewDocument("c:\temp\Assembly.asmdot", 0, swSheetWidth,
swSheetHeight)
Dim swAssembly As SldWorks.AssemblyDoc
Set swAssembly = Part
swApp.ActivateDoc2 "Assem3", False, longstatus
Set Part = swApp.ActiveDoc
Set myModelView = Part.ActiveView
myModelView.FrameState = swWindowState_e.swWindowMaximized

' Insert Component
Dim AssemblyTitle As String
AssemblyTitle = Part.GetTitle
Dim tmpObj As SldWorks.ModelDoc2

```

```

Dim errors As Long
Set tmpObj = swApp.OpenDoc6("c:\temp\Cylinder_225.SLDPRT", 1, 32, "", errors,
longwarnings)
Set Part = swApp.ActivateDoc3(AssemblyTitle, True, 0, errors)
Dim swInsertedComponent As SldWorks.Component2
Set swInsertedComponent = Part.AddComponent5("c:\temp\Cylinder_225.SLDPRT", 0,
"", False, "", 0.721748083354866, 0.92923566513213, 1.36775956940221)
swApp.CloseDoc "c:\temp\Cylinder_225.SLDPRT"

boolstatus = Part.Extension.SelectByID2("Cylinder_225-1", "COMPONENT", 0, 0, 0,
False, 0, Nothing, 0)
Part.UnfixComponent
' Create PLANE1 and PLANE2

boolstatus =
Part.SetUserPreferenceToggle(swUserPreferenceToggle_e.swDisplayPlanes, True)
boolstatus = Part.Extension.SelectByID2("Front Plane", "PLANE", 0, 0, 0, True, 0,
Nothing, 0)
Dim myRefPlane As SldWorks.RefPlane
Set myRefPlane = Part.FeatureManager.InsertRefPlane(8, 0.1, 0, 0, 0, 0)
boolstatus = Part.Extension.SelectByID2("Front Plane", "PLANE", 0, 0, 0, True, 0,
Nothing, 0)
Set myRefPlane = Part.FeatureManager.InsertRefPlane(8, 0.001, 0, 0, 0, 0)
Part.ClearSelection2 True

' Insert Component
AssemblyTitle = Part.GetTitle
Set tmpObj = swApp.OpenDoc6("c:\temp\Cylinder_225.SLDPRT", 1, 32, "", errors,
longwarnings)
Set Part = swApp.ActivateDoc3(AssemblyTitle, True, 0, errors)
Set swInsertedComponent = Part.AddComponent5("c:\temp\Cylinder_225.SLDPRT", 0,
"", False, "", -9.24876597127877E-02, 5.65812429995276E-02, 0.22909678472206)
swApp.CloseDoc "c:\temp\Cylinder_225.SLDPRT"

Part.ClearSelection2 True

boolstatus = Part.Extension.SelectByID2("PLANE1", "PLANE", 0.012624285665936,
0.154967158689601, 0.112309569516221, True, 1, Nothing, 0)
boolstatus = Part.Extension.SelectByRay(-0.076247080467482, 0.103507691590153,
0.115945777078347, -0.283810108456217, -0.958610200111282, 2.27663475480263E-02,
2.30409955137483E-03, 1, True, 1, 0)

' Create CoincidentMateFeatureData
Dim MateData As SldWorks.CoincidentMateFeatureData
Set MateData = Part.CreateMateData(0)

' Set the Entities To Mate
Dim EntitiesToMate(1) As Object
Set EntitiesToMate(0) = Part.SelectionManager.GetSelectedObject6(1, -1)

```

```

Set EntitiesToMate(1) = Part.SelectionManager.GetSelectedObject6(2, -1)
Dim EntitiesToMateVar As Variant
EntitiesToMateVar = EntitiesToMate
MateData.EntitiesToMate = (EntitiesToMateVar)

' Set the Mate Alignment
MateData.MateAlignment = 1

' Create the mate
Dim MateFeature As SldWorks.Feature
Set MateFeature = Part.CreateMate(MateData)
Part.ClearSelection2 True
Part.EditRebuild3

Part.ClearSelection2 True
boolstatus = Part.Extension.SelectByID2("PLANE2", "PLANE", -0.051265823820458,
0.210240836185798, -8.14719342065839E-02, True, 1, Nothing, 0)
boolstatus = Part.Extension.SelectByRay(0.767058330396253, 0.909200270895155,
1.48052500817425, -0.400036026779312, -0.515038074910024, -0.758094294050284,
1.12587739183236E-02, 1, True, 0, 0)

' Create CoincidentMateFeatureData
Set MateData = Part.CreateMateData(0)

' Set the Entities To Mate
Set EntitiesToMate(0) = Part.SelectionManager.GetSelectedObject6(1, -1)
Set EntitiesToMate(1) = Part.SelectionManager.GetSelectedObject6(2, -1)
EntitiesToMateVar = EntitiesToMate
MateData.EntitiesToMate = (EntitiesToMateVar)

' Set the Mate Alignment
MateData.MateAlignment = 1

' Create the mate
Set MateFeature = Part.CreateMate(MateData)
Part.ClearSelection2 True
Part.EditRebuild3
Part.ViewZoomtofit2

Stop ' Delete Plane2; create a new plane that is 1mm from the Front Plane;
select the Mates folder in the FeatureManager design tree; then press F5 to repair
the broken mate

Dim swModel As SldWorks.ModelDoc2
Dim cnt As Long

```

```

Set swModel = swApp.ActiveDoc

Dim swAsm As SldWorks.AssemblyDoc
Set swAsm = swModel

Dim failedMates As Variant
Dim ProcessedMates As Variant
Dim lStatus As Long
lStatus = swAsm.AutoMateRepair(ProcessedMates, failedMates)

Debug.Print "Number of Processed Mates is " & UBound(ProcessedMates) + 1 & ":"
For cnt = 0 To UBound(ProcessedMates)
Debug.Print ProcessedMates(cnt).Name
Next

Debug.Print vbNewLine

Debug.Print "Number of failed Mates is " & UBound(failedMates) + 1 & ":"
For cnt = 0 To UBound(failedMates)
Debug.Print failedMates(cnt).Name
Next
End Sub

```

#### Remarks

Before calling this method you must:

- ensure that the entities needed to repair the broken mates are present in the assembly
- and -
- select the Mates folder in the FeatureManager design tree.

#### See Also

[IAssemblyDoc Interface](#)  
[IAssemblyDoc Members](#)  
[IAssemblyDoc::CreateMate Method \(\)](#)

#### Availability

SOLIDWORKS 2025 SP01, Revision Number 33.1

SOLIDWORKS API Help

[Send comments on this topic.](#)

### ClearanceVerificationManager Property (IAssemblyDoc)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IAssemblyDoc Interface](#) : ClearanceVerificationManager Property (IAssemblyDoc)

Gets the clearance verification manager.

#### .NET Syntax

Visual Basic (Declaration)

```
ReadOnly Property ClearanceVerificationManager As System.Object
```

```
Dim instance As IAssemblyDoc
Dim value As System.Object

value = instance.ClearanceVerificationManager
```

C#

```
System.Object ClearanceVerificationManager {get;}
```

C++/CLI

```
property System.Object^ ClearanceVerificationManager {
    System.Object^ get();
}
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Property Value

[IClearanceVerificationMgr](#)

### Visual Basic for Applications (VBA) Syntax

See [AssemblyDoc::ClearanceVerificationManager](#).

### Example

See the [IClearanceVerificationMgr](#) examples.

### See Also

[IAssemblyDoc Interface](#)  
[IAssemblyDoc Members](#)  
[IClearanceResult Interface](#)

### Availability

SOLIDWORKS 2024 SP01, Revision Number 32.1

SOLIDWORKS API Help

[Send comments on this topic.](#)

## CollectAllSupportiveMates Method (IAssemblyDoc)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IAssemblyDoc Interface](#) : CollectAllSupportiveMates Method (IAssemblyDoc)

Gets all mates supportive of a mate controller in this assembly.

### .NET Syntax

Visual Basic (Declaration)

```
Function CollectAllSupportiveMates() As System.Object
```

Visual Basic (Usage)

```
Dim instance As IAssemblyDoc
Dim value As System.Object

value = instance.CollectAllSupportiveMates()
```

C#

```
System.Object CollectAllSupportiveMates()
```

C++/CLI

```
System.Object^ CollectAllSupportiveMates();
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Return Value

Array of supportive [mates](#)

### Visual Basic for Applications (VBA) Syntax

See [AssemblyDoc::CollectAllSupportiveMates](#).

### Example

See the [IMateControllerFeatureData](#) examples.

### Remarks

This method is valid only if [IAssemblyDoc::IsSupportedMatesAvailable](#) is true.

To create a mate controller, see the Remarks section of [IMateControllerFeatureData](#).

For more information about the Mate Controller feature, see the **SOLIDWORKS Design user-interface help > Assemblies > Mates > Mate Controller** topic.

### See Also

[IAssemblyDoc Interface](#)  
[IAssemblyDoc Members](#)

### Availability

SOLIDWORKS 2023 FCS, Revision Number 31

SOLIDWORKS API Help

[Send comments](#) on this topic.

## CompConfigProperties Method (IAssemblyDoc)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IAssemblyDoc Interface](#) : CompConfigProperties Method (IAssemblyDoc)

**Obsolete. Superseded by [IAssemblyDoc::CompConfigProperties4](#).**

### .NET Syntax

Visual Basic (Declaration)

```
Sub CompConfigProperties( _  
    ByVal M\_suppressed As System.Boolean, _  
    ByVal M\_show\_component As System.Boolean, _  
    ByVal M\_fdetail As System.Boolean _  
)
```

Visual Basic (Usage)



```
Dim instance As IAssemblyDoc  
Dim M_suppressed As System.Boolean  
Dim M_show_component As System.Boolean  
Dim M_fdetail As System.Boolean  
  
instance.CompConfigProperties(M_suppressed, M_show_component, M_fdetail)
```

C#

```
void CompConfigProperties(  
    System.bool M\_suppressed,  
    System.bool M\_show\_component,  
    System.bool M\_fdetail  
)
```

C++/CLI

```

void CompConfigProperties(
& System.bool M\_suppressed,
& System.bool M\_show\_component,
& System.bool M\_fdetail
)

```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Parameters

*M\_suppressed*

*M\_show\_component*

*M\_fdetail*

### Visual Basic for Applications (VBA) Syntax

See [AssemblyDoc::CompConfigProperties](#).

### See Also

[IAssemblyDoc Interface](#)  
[IAssemblyDoc Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

## CompConfigProperties2 Method (IAssemblyDoc)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IAssemblyDoc Interface](#) : CompConfigProperties2 Method (IAssemblyDoc)

Obsolete. Superseded by [IAssemblyDoc::CompConfigProperties4](#).

### .NET Syntax

#### Visual Basic (Declaration)

```

Function CompConfigProperties2( _
    ByVal Suppression As System.Integer, _
    ByVal Visibility As System.Boolean, _
    ByVal FeatureDetails As System.Boolean _
) As System.Boolean

```

#### Visual Basic (Usage)



```

Dim instance As IAssemblyDoc
Dim Suppression As System.Integer
Dim Visibility As System.Boolean
Dim FeatureDetails As System.Boolean
Dim value As System.Boolean

```

```
value = instance.CompConfigProperties2(Suppression, Visibility, FeatureDetails)
```

#### C#

```

System.bool CompConfigProperties2(
    System.int Suppression,
    System.bool Visibility,
    System.bool FeatureDetails
)

```

#### C++/CLI

```

System.bool CompConfigProperties2(
& System.int Suppression,
& System.bool Visibility,
& System.bool FeatureDetails
)

```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

## Parameters

*Suppression*

*Visibility*

*FeatureDetails*

## Visual Basic for Applications (VBA) Syntax

See [AssemblyDoc::CompConfigProperties2](#).

## See Also

[IAssemblyDoc Interface](#)

[IAssemblyDoc Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

## CompConfigProperties3 Method (IAssemblyDoc)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IAssemblyDoc Interface](#) : CompConfigProperties3 Method (IAssemblyDoc)

Obsolete. Superseded by [IAssemblyDoc::CompConfigProperties4](#).

## .NET Syntax

### Visual Basic (Declaration)

```
Function CompConfigProperties3( _  
    ByVal Suppression As System.Integer, _  
    ByVal Solving As System.Integer, _  
    ByVal Visibility As System.Boolean, _  
    ByVal FeatureDetails As System.Boolean _  
) As System.Boolean
```

### Visual Basic (Usage)



```
Dim instance As IAssemblyDoc  
Dim Suppression As System.Integer  
Dim Solving As System.Integer  
Dim Visibility As System.Boolean  
Dim FeatureDetails As System.Boolean  
Dim value As System.Boolean  
  
value = instance.CompConfigProperties3(Suppression, Solving, Visibility, FeatureDetails)
```

### C#

```
System.bool CompConfigProperties3(  
    System.int Suppression,  
    System.int Solving,  
    System.bool Visibility,  
    System.bool FeatureDetails  
)
```

### C++/CLI

```
System.bool CompConfigProperties3(  
& System.int Suppression,  
& System.int Solving,  
& System.bool Visibility,  
& System.bool FeatureDetails  
)
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

## Parameters

*Suppression*

*Solving*

*Visibility*

*FeatureDetails*

## Visual Basic for Applications (VBA) Syntax

See [AssemblyDoc::CompConfigProperties3](#).

## See Also

[IAssemblyDoc Interface](#)  
[IAssemblyDoc Members](#)